GENERAL — P&ID LEGEND, SYMBOLS, LINE NUMBERING AND CONVENTIONS

Applies to all SEVERO WM OIL P&IDs (GEN / PS-01 / PS-02 / PS-03 / PS-06). Symbols per ISA 5.1 and ISO 10628 practice, adapted. Conventions agreed 17/18-Oct-2026 (G-6).

VALVES AND IN-LINE PIPING ITEMS

- Gate valve — ISOLATION, normally open (N.O.)
- Gate valve — normally closed (N.C.), filled
- Globe valve — THROTTLING (manual)
- Globe valve — normally closed (N.C.)
- Knife-gate valve — sludge / solids service
- Angle globe — manual back-pressure (HV-253, HV-303)
- Check valve (NRV) — flow left to right
- PSV — inlet from below, outlet to the side
- Spectacle blind (open half in line)
- Reducer: concentric / eccentric flat-on-top
- Flexible connector (pump nozzles)
- Simplex basket strainer
- Duplex basket strainer (cleanable on line)
- Chemical injection quill
- Bellmouth suction intake (facing down)
- Drain / vent ¾": gate valve + cap
- Sample cock + funnel
- Open tundish to drain (visible discharge) — A1 C1
- Piping class break
- Insulated line (dashed line alongside)

CSO / CSC = car-sealed open / closed; LO / LC = locked open / closed.
HV-NNN above a valve = operating line-up valve (listed in the valve list).

LINE TYPES AND SERVICE CODES

- P process oil (wet / dry) — main lines
- PR product to storage / send-out
- W water / effluent
- SL sludge
- HO hot oil (thermal fluid)
- VA vapour / vacuum
- CWS / CWR cooling water supply / return
- CH chemical (demulsifier)
- DS diesel / solar
- DR / VT drains, slop, vents
- Standby / future equipment and lines

Colours are reading aids only: the service is defined by the line number.

ABBREVIATIONS

BL	battery limit	**LCS**	local control station (at the motor)
CT	cone tip (PS-02 elevation datum)	**MCC**	motor control centre
DOL	direct-on-line motor starter	**NPS**	nominal pipe size, inch
EL	elevation, m	**NRV**	non-return (check) valve
FOT	flat on top (eccentric reducer)	**PD**	positive displacement
IFR	issued for review	**TQ**	technical query to the Owner

EQUIPMENT

- **Centrifugal pump** + motor (M)
- **Rotary lobe pump** (positive displacement)
- **Standby pump** (dashed outline)
- **Liquid-ring vacuum pump** (VP-301)
- **Shell-and-tube** exchanger
- **Fan (exhaust / CT)** EF-101, CT fan
- **Elevated cone-bottom tank,** cone roof (PS-02)
- **Atmospheric flat-bottom tank, cone** roof (PS-06, TK-801)
- **Vertical vessel,** top head (V-301)
- **Cooling tower,** induced draft (CT-801)
- **Vendor package** (skid boundary) — X-201
- **Off-page** connector — to Sheet 06, line no. / item
- **HOLD cloud: pending** Owner data (TQ-004) — HOLD
- **Revision triangle** (change in that revision) — 1

22 kW [A] — [A] = Assumed value, to be confirmed by Vendor / Owner.

INSTRUMENT SYMBOLS AND SIGNALS (ISA 5.1)

- **Field-mounted** local instrument — PG 202
- **On local control** station (LCS) at motor — HS 103A
- **In MCC / electrical** panel (trip relay, alarm) — XA 201
- **Interlock / trip** (see TRIP SCHEDULE) — I-101
- **Variable-frequency** drive (P-202 only) — VFD
- **Electric motor** (kW in MOTOR DATA) — M

- Process connection / impulse line
- Electrical signal (hard-wired)
- Pneumatic signal (future, if plant goes semi-auto)
- Capillary (filled system)

STANDARD INSTRUMENT HOOK-UPS

PG ½" root gate valve; pulsation snubber on PD-pump lines.
TI dial thermometer in a 1" flanged thermowell (TW).
LI radar level on a roof nozzle, local indication only.
LG gauge glass with gate isolation top and bottom.
LSLL / LSH level switch hard-wired to the MCC trip relay.
GD fixed gas detector LEL / H2S / O2, alarm at the gallery entrance and in the MCC.
Every instrument process connection has its own root valve.
All trip switches wired FAIL-SAFE: NC contact, de-energise to trip (wire break / loss of supply stops the motor; IEC 61511). 1

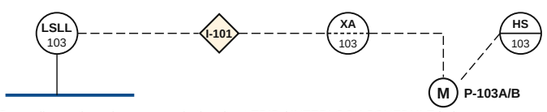
INSTRUMENT LETTERS (ISA 5.1)

1st	Measured variable / modifier	Succ.	Function / modifier
A	Analysis (gas: LEL / H2S / O2)	A	Alarm
F	Flow	E	Primary element (sensor)
H	Hand (manual)	G	Gauge / glass (local reading)
L	Level	I	Indicate
P	Pressure	S	Switch
S	Speed (VFD)	T	Transmitter
T	Temperature	W	Thermowell
V	Vibration	C	Control (SC speed set)
X	Unclassified (trip relay)	H / L	High / Low
D	(after P: differential, PDG)	HH / LL	High-high / Low-low
Q	(after F: totalise, FQI)		

Tag = letters + number; 1st digit = unit (LSLL-103 = PS-01). Parallel numbering: numbers are unique per 1st letter (PG-110 ≠ TI-110); HV hand valves, I-xxx interlocks and equipment tags each run their own series.
Succ. = succeeding letters (function / modifier). D and Q modify the 1st letter.

HOW A TRIP IS SHOWN (EXAMPLE: P-103 DRY-RUN PROTECTION)

Switch → interlock diamond → MCC trip relay → motor stop. Start / stop by HS at the LCS.



Same diamond number appears in the sheet TRIP / INTERLOCK SCHEDULE below.

TRIP / INTERLOCK SCHEDULE — format used on every unit sheet

I-No.	Initiator	Setpoint	Action	Reset
I-101	LSLL-103 (C2)	EL −3.10 m	Stop P-103A/B	Manual, LCS

MOTOR DATA — format used on every unit sheet

Tag	kW	Starter	Control	Trip by
P-202A/B	55	VFD	LCS + speed set	local stop
P-103A/B	22 [A]	DOL	LCS start/stop	I-101

GENERAL NOTES

1. All valves hand-operated: GATE = isolation, GLOBE = throttling, KNIFE-GATE = sludge / solids. NRV on every pump discharge.
2. Every positive-displacement pump: discharge PSV back to its suction source; set point on the sheet.
3. Automatic actions = equipment protection only. Each is shown by an interlock diamond and listed in the TRIP SCHEDULE; motors in MOTOR DATA. Electrical & control scope is readable from the sheets.
4. Units: NPS inch; m; °C; pressure written as bar(a) or bar(g).
5. Valve positions = normal operation (N.O. open / N.C. filled).
6. ¾" vents / drains shown at equipment and key points; the full set is added with the piping isometrics (Detailed Engineering).
7. Schematic, not to scale; elevations in m from the stated datum.

VALVE TYPE BY SERVICE

Service	Valve
Isolation ≥ 2" (oil, water, CW, diesel)	Gate, CS, flanged Cl.150
Isolation ≤ 1½" (drain, vent, root)	Gate, forged CS
Throttling / back-pressure / bypass	Globe (angle globe at BPV)
Sludge, tank-bottom drain N4	Knife-gate, bi-directional
Pump discharge	Check: swing / dual-plate
Hot oil (class B1)	HOLD: bellows-sealed [A]
Vacuum (class V1)	Gate / globe, vacuum packing
C1 → C2 U-link (PS-01)	Double block & bleed, LC

REFERENCED CODES & STANDARDS

Code	Applied to
ANSI/ISA-5.1	Instrument identification and symbols
ISO 10628-2	Equipment / piping graphical symbols
ASME B16.5 / B16.9 / B16.10	Flanges Cl.150; fittings; valve F-F
API 650 / API 620 [A] / UL-142 [A]	Storage tanks (PS-06, TK-801 / PS-02)
API 2000	Tank and pit venting
API RP 2003	Static electricity, fill velocity
API 676	Rotary lobe (PD) pumps
ANSI/HI 9.8	Pump intake design, submergence
TEMA	Shell-and-tube exchangers
NFPA 30	Class IIIB liquids, spacing, bunds
IEC 61511	Fail-safe protective functions
ACI 350 / EN 1992-3	Liquid-retaining concrete
EN 476 / EN 124	Man-entry openings; cover load class
ISO 14122-4	Fixed ladders (gallery)
ASTM D4007 / D473	Water & sediment tests (TQ)

LINE NUMBER FORMAT 1

8" - P - 2105 - A1 - IH

size (NPS) | service code | sequence no. | class | insulation

Sequence no.: 4 digits = unit digit + serial (2105 = PS-02, line 105).
Service code: see LINE TYPES AND SERVICE CODES. Line service codes are independent of ISA instrument letters (P = process oil, not pressure).

PIPING CLASSES (preliminary)

Class	Service / basis
A1	CS, ASME B16.5 Cl.150 RF; process ≤ 100 °C (oil, water,
	sludge, product, diesel, drains)
B1	Hot oil (thermal fluid) — HOLD, pending TQ-004
V1	PS-03 vapour / vacuum, CS, full-vacuum design
C1	Cooling water (CWS / CWR), CS Cl.150
S1	Chemical dosing, SS316L, small bore [A]

INSULATION CODES

Code	Meaning
IH	Heat conservation, rock wool 50 mm + Al cladding
IH75	Heat conservation 75 mm (E-201 shell, hot-oil lines)
PP	Personnel protection (surface > 60 °C)
(none)	Not insulated
	CUI coating under all insulation (D-W).

UNIT CODES (1st digit of line / instrument numbers)

Code	Unit
1	PS-01 Receiving pit + dry gallery
2	PS-02 Heated settling tanks T-201…T-204
3	PS-03 Main separator + vacuum + chiller
6	PS-06 Product storage + send-out
8	Utilities: cooling tower CT-801, diesel TK-801
9	Effluent TK-901 + sludge

EQUIPMENT TAG PREFIXES

Tag	Equipment
P / VP	Pump / vacuum pump
E	Heat exchanger
V / T / TK	Vessel / tank / pit
ST / X / BSK / SK	Strainer / package / basket / skip
CT / CH	Cooling tower / chiller
EF / GD / WP / RH	Exh. fan / gas det. / wall piece / roof hatch

Rule: prefix-UNN (U = unit). Renumbered 18-Oct: T-21…24 → T-201…204; PS-03 E-1 → E-301, P-3A/B/C → P-301A/B/C, VP-1 → VP-301, CH-1 → CH-301; TK-DSL → TK-801.

DRAWING REGISTER — this PDF holds ALL project drawings (22 pages)

Pg	Sh.	Unit	Content
1	**01**	General	**Legend, symbols and conventions**
2–3	PFD	General	Process flow diagram GEN-PFD-001 Rev P4 (2 sheets: PFD + vessel / line schedule)
4	02	General	Interconnection diagram (all units)
5	03	PS-01	Tanker unloading and receiving pit
6	03A	PS-01	Receiving pit details: falls, sections, U-link
7	04	PS-01	Dry pump gallery P-101 / P-102 / P-103
8	05	PS-02	Settling tanks T-201 … T-204
9	06	PS-02	Fill header, X-201 dosing, ST-201
10	07	PS-02	Heater E-201, P-202, transfer P-201
11	08	PS-03	Separator V-301, heater loop, P-301A/B/C
12	09	PS-03	Vacuum and condensing train
13	10	PS-03	VP-301 seal-water loop and chiller CH-301
14	11	PS-06	Product storage and send-out
15	12	Utilities	Cooling tower CT-801 and CW network
16	13	Utilities	Diesel storage TK-801 and transfer
17	14	Effluent	Effluent TK-901 and sludge handling
18	GA	General	Site GA / plot plan GEN-LAY-001 Rev P1
19–21	LAY	PS-01	Pit + dry gallery GA PS01-LAY-001 Rev.2 (3 sheets)
22	LAY	PS-03	V-301 internals GA PS03-LAY-001 Rev P5

REVISION HISTORY

Rev	Date	Status	Description
P0	18-Oct-2026	IFR	First issue
P1	19-Oct-2026	IFR	M-01…M-20; Sh. 03A; GA; register; PFD P4 + LAY P5 aligned

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 01 — Legend, symbols and conventions

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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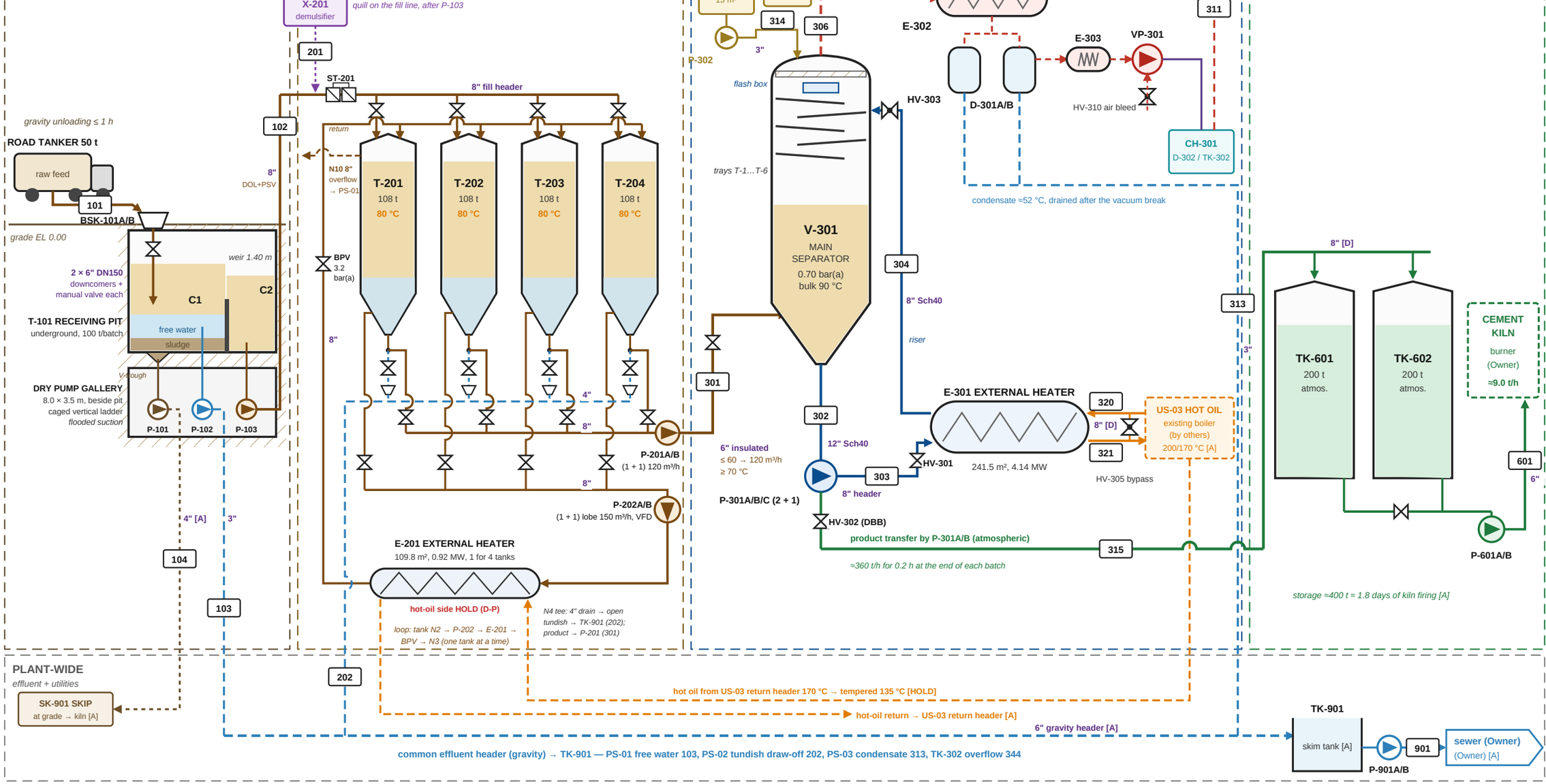
Ref.: Doc. Control Legend Rev1; PS01/02/03 DB; G-6, G-7

Date: 19-Oct-2026

Sheet 01 of 15

A3 — NTS (schematic)

Supersedes Rev P0 (18-Oct-2026)



Stream No.	101	102	103	104	201	202	301	306	313	311	314	315	601	901	320/321	330/331
Stream	**Raw feed** tanker→T-101	**PS-01 oil** → PS-02	**PS-01 free** water	**PS-01** sludge	**Demulsifier** → line 102	**PS-02 cone** draw-off	**Wet charge** → V-301	**Vapour** V-301→E-302	**Condensate** → effluent	**Vent gas** VP-301 → atm	**Diesel /** solar → V-301	**Product** → PS-06	**Fuel to** kiln	**Effluent to** drainage	**Hot oil** E-301 (US-03)	**Cooling w.** E-302 (CT-801)
Phase	L (W/O)	L (W/O)	L (brine)	slurry	L	L + solids	L (W/O)	V	L	G	L	L	L	L	L	L
T (°C)	10-38	10-38	10-38	ambient	ambient	≈80	≥ 70	89.9-98.5	≈52	19-20	30 [A]	≈97	HOLD	≈49 [A]	200 → 170	32 → 47.8
P (bar(a))	1.01	1.01	1.01	1.01	—	1.01	1.01	0.70	1.01	1.01	1.01	1.01	—	1.01	[A]	[A]
Flow (t/day)	407.4	325.0	82.4	TBD [A]	0.003-0.010	55.8	269.1	66.6	66.6	1.40	14.0	216.5	216.5	≈205	—	—
per batch (t)	—	—	—	—	—	—	89.713	22.2	22.207	0.468	4.667	72.158	—	—	55.7 GJ	—
Oil/water/salt wt%	49.1/49.1/1.8	61.5/37.1/1.4	0/96.4/3.6	TBD	—	0/96.4/3.6	74.3/24.8/0.9	H2O + NC	H2O, trace HC	air + H2O	diesel	92.4/≈0/1.1 *	as 315	0/97.6/2.4	thermal oil	water
Rate	75 m³/h	80 m³/h	≥ 40 m³/h	20-30 m³/h [A]	10-30 ppm	≈140 m³/h	≤60 → 120 m³/h	6.53 t/h	batch	≈100 kg/h	25 m³/h	≈360 t/h	≈9.0 t/h	P-901	249 m³/h	238 m³/h
Status	[A]	R2	[A]	HOLD	HOLD	P1	P18	P18	P18	P18	P18	P18	HOLD	[A]	P18	P18
Line size	**hose [A]**	**8"**	**3"**	**4" [A]**	**½" [A]**	**4"**	**6"**	**DN450**	**3"**	**3" [A]**	**3"**	**8"**	**6" (G-1)**	**3" [A]**	**8" [D]**	**8" [D]**

STREAM TABLE — design case (raw feed 50 wt% water), preliminary plant water balance. * plus diesel 6.47 wt% (7 % of the dry oil [A]). [A] = assumed, to be confirmed (TQ-004). R2 = PS01-DB-001 Rev.2; P1 = PS02-DB-001 Rev P1; P18 = PS03-DB-001 Rev P18 Section 3 (HMB). HOLD = not yet closed.

SEVERO WM OIL — OVERALL PFD — SHEET 2: VESSEL & EQUIPMENT SCHEDULE (DIMENSIONS, THICKNESS, WEIGHTS) AND LINE SCHEDULE

CONCEPT / IFR — preliminary engineering. Tags and line numbers per P&ID GEN-PID-001 Rev P1. Dimensions per GEN-EL-001 Rev P2; PS01-DB-001 Rev.2; PS02-DB-001 Rev P1; PS03-DB-001 Rev P18. Weights = steel only (no supports, foundations, insulation, platforms).

A. VESSEL & EQUIPMENT SCHEDULE — metallic items (dimensions per GEN-EL-001 Rev P2; thickness: DB value where screened, otherwise [A])

No	Tag	Service	Shape / type	Qty	Diameter	Height / length	Volume	Design P / T	Material	Shell / cone / head (mm)	Empty t	Oper. t	Hydro t	Status
PS-01 RAW FEED RECEPTION — PS01-DB-001 Rev.2 / LAY-001 Rev.2 (19-Oct-2026)														
1	BSK-101A/B	Inlet coarse baskets	perforated box in HOP-101 collar	2	0.80 × 0.80 m	0.45 m deep	—	atm / amb.	CS [A]	6 [A] perf. Ø12 @16	0.08	0.38	—	CURRENT
2	HOP-101	Receiving hopper (M-04)	collar + pyramid + neck, fixed	1	1.70 × 1.00 m top	0.50 + 0.95 + 0.15 m	1.67 m³	atm; uplift 7.7 kN	CS [A]	8 [A]	0.49 [D]	2.0	2.2	CURRENT
PS-02 HEATED SETTLING TANKS — PS02-DB-001 Rev P1 (hot-oil side of E-201 HOLD)														
3	T-201…T-204	Heated settling tanks, 108.3 t	vertical, 30° cone, cone roof, 8 legs	4	7.00 m	3.08 cyl + 2.02 cone	144.4 m³ (125 work.)	atm / 80 °C	CS (API 620 cone) [A]	6 / 8 / 6 roof [A]	8.6	117	153	CURRENT
4	E-201	PS-02 external heater (1 for 4)	shell & tube, 6 tube passes	1	shell ≈1.1 m [D]	tubes 9.66 m; 2" × 60	109.8 m²	≤ 6.0 bar(a) tube	CS	10 shell [A]	8.3	14.2	15.9	CURRENT
PS-03 MAIN SEPARATOR — PS03-DB-001 Rev P18 (frozen for print)														
5	V-301	Main separator (vacuum)	vertical, cone bottom, F&D head	1	5.294 m	7.704 + 2.217 + 0.896 m	≈198 m³	5.0 bar(g) & FV / 150 °C	SA-516-70	25 / 25 [A] / 20 (FV)	46.0	134.9	244.0	FROZEN
6	E-301	External process heater	S&T TEMA AES/AET, 6 passes	1	shell ≈1.27-1.43 m [D]	tubes 9.66 m; 2" × 132	241.5 m²	tube 8 / shell 6.5 bar(g)	CS	12 shell [A]	15.6	25.8	28.5	FROZEN
7	E-302	Main surface condenser	TEMA E-shell, inclined	1	shell ID 0.65 m	tubes 6.0 m; 1" × 250	120 m² + SC	5.0 bar(g) & FV	CS	10 (DB)	3.7	4.7	5.8	FROZEN
8	D-301A/B	Condensate receivers	drum, 2:1 heads, CW jacket	2	2.2 m [A]	2.48 m T/T [D]	12.2 m³	5.0 bar(g) & FV	CS	10 + 6 jacket [A]	3.1	13.1	15.3	FROZEN
9	E-303	Vent condenser	small S&T, above D-301A/B	1	vendor	vendor	≈2.3 m²	5.0 bar(g) & FV	SS316L [A]	vendor	≈0.3 [A]	≈0.35	—	vendor
10	D-302	VP-301 discharge K.O. drum	vertical gravity drum, atm.	1	0.30 m	1.0 m T/T [A]	0.07 m³	atm	SS316L / PP [A]	4 [A]	0.05	0.06	0.12	FROZEN
11	TK-301	Diesel / solar day tank	vertical, flat bottom, atm.	1	2.5 m [D]	3.2 m [D]	15 m³	atm / amb.	CS [A]	6 / — / 6 [A]	1.8	14.4	17.5	[A]
PS-06 PRODUCT STORAGE — closed 16-Oct-2026 (2 × 200 t, H/D 1.3)														
12	TK-601/602	Product storage tanks, 200 t	vertical, flat bottom, cone roof	2	6.4 m	8.3 m (H/D 1.3)	267 m³	atm / ≤ 97 °C	CS A283-C	6 (API 650) / 8 / 6 [A]	12.6	212.6	279.6	FROZEN
UTILITIES — diesel bulk storage (G-4, P&ID Sheet 13)														
13	TK-801	Diesel / solar bulk tank (≥ 2 d)	vertical, cone roof, bunded 9 × 9 m	1	3.6 m [A]	5.0 m [A]	50.9 m³	atm / amb.	CS	6 / 8 / 6 [A]	4.2	43	55	CURRENT

TOTAL STEEL, all metallic items (empty, all quantities) ≈ 146 t — 13 tags, 19 units. Heaviest lift V-301 ≈ 46 t; heaviest operating load TK-601/602 ≈ 213 t each; hydrotest governs foundations. HOP-101 0.49 t [D] = 7.0 m² × 8 mm × 7,850 kg/m³ × 1.10.

A2. NON-METALLIC / CIVIL ITEMS (not in the steel total)

Tag	Item	Dimensions	Key data	Material	Status
T-101	PS-01 receiving pit (RC)	8.0 × 5.0 m internal; C1 5.6 + C2 2.4 m; mean floor EL −3.30; C1 floor 1:20 (−3.19 → −3.41); trough 1:50 to sump −4.51	weir 1.40 m (crest −1.90); 134.9 m³ to grade, 120.9 m³ to LAH-101; batch EL −0.52	RC (ACI 350 / EN 1992-3)	CURRENT Rev.2
—	PS-01 dry pump gallery (RC)	8.0 × 3.5 m internal, floor EL −3.80, falls 1:100 to sump 0.6 × 0.6 × 0.6 m	monolithic with T-101; 9 cast-in wall pieces; hatches 800 × 800 (pit)	RC	CURRENT Rev.2
TK-901	Effluent cool-down + oil-skim tank	below grade, 2 compartments, ≈ 80 m³ working (55 + 25) [A]	cool-water heel ≥ 53 m³; skimmer + P-902; P-901A/B to sewer	RC, oil-tight lining	CURRENT [A]
TK-302	VP-301 seal-water return tank	Ø0.7 × 0.9 m; 0.23 m³ working; overflow DN40 at +0.60 m	open to atmosphere; insulated	HDPE / PP / FRP	FROZEN
Bunds	PS-02 / PS-06 / TK-801	PS-02 21.2 × 21.2 m; PS-06 21.8 × 12.4 m (wall ≈1.6 m [A]); TK-801 9 × 9 m	≥ 110 % of the largest tank (137.5 / 294 / 56 m³)	RC	GA GEN-LAY-001 P1

B. LINE SCHEDULE — stream / P&ID line number (GEN-PID-001 Rev P1), nominal size Sch40 unless stated

Stream	P&ID line No.	From → To	Fluid	Size	Design flow	Velocity	Main manual valves	Basis
101	6"-P-1101/1102-A1	tanker → BSK-101A/B → HOP-101 → DC-101A/B → T-101 C1	raw wet oil	hose [A] / 2 × 6"	75 m³/h each	≈1.1 m/s	HV-101A/B N.O. (spindle to grade)	PS01-DB §6-8
102 s	10"-P-1108 / 1109-A1	C2 pocket → WP-103A/B → P-103A/B (one line per pump)	raw wet oil, cold	10" each	80 m³/h	0.44 m/s	HV-113A/B gate; ST-103A/B basket	PS01-DB §11.2
102	8"-P-1113/1114 → 1115, 2101	P-103A/B → X-201 → ST-201 → fill header → T-201…T-204 N1	wet oil + demulsifier	8"	80 m³/h	0.69 m/s	PSV + NRV + HV-114A/B; ST-201; HV-2k1	PS01 Rev.2 §11.3; PS02 P1
103	4"-W-1105…1107 / 3"-W-1122	WP-102a/b/c → P-102A/B → TK-901 (during unloading too)	free water (brine)	4" draw-off / 3"	≥ 40 m³/h	2.3 m/s (3")	HV-107a/b/c + cocks; HV-110 throttle; HV-111	PS01 Rev.2 §10.2 (M-12)
rag	3"-P-1123 → 4"-P-1110	P-102 → HV-112 → return header WPR → C2	rag / emulsion	3" / 4"	≤ 40 m³/h	—	HV-112 N.C.	PS01 Rev.2 §10.3
104	6"-SL-1103 / 4"-SL-1120 [A]	trough sump → dip pipe → WP-101 → P-101A/B → SK-901	sludge	6" suct. / 4"	20-30 m³/h [A]	≤ 1.1 m/s	HV-103 knife; HV-119 knife; SP-103	PS01 Rev.2 §10.1
104 w	4"-W-1126-A1	P-101 → HV-118 → TK-901 (residual bottom water)	water	4"	20-30 m³/h [A]	—	HV-118 N.C.	M-13
104 r	4"-P-1128-A1	P-101 → HV-120 → WPR → C2 (C1 residual ≈ 7.8 m³/batch)	wet oil	4"	20-30 m³/h [A]	—	HV-120 N.C.	M-20
U-link	6"-P-1124-A1	T-101 C1 (WP-104a) ↔ C2 (WP-104b), U-loop in the gallery	wet oil (gravity)	6"	≈ 89 m³/h at 10 °C	—	DBB HV-115 knife + HV-116 gate (LC); SP-102; HV-117	M-13 / M-17
N10	8"-P-2110-A1	T-201…T-204 overflow → T-101 C1 (gravity, no valve)	wet oil	8"	≤ 80 m³/h	part-full	none (must stay open)	PS02 P1 (D-V)
loop	8"-P-2115 / 2120-A1-IH	T-20x N2 → P-202A/B → E-201 → BPV → N3	wet oil 20 → 80 °C	8"	117 (150 rated)	1.0-1.3 m/s	HV-2k2 / 2k3; BPV HV-253 + PG-277	PS02 P1
202	4"-W-214k / 6"-W-2145-A1	T-20x N4 → drain HV → tundish → TK-901	hot water + fines	4" / 6" hdr	≈ 140 m³/h	gravity	HV-2k6 per tank	PS02 P1
prod.	6"-P-213k / 8"-P-2135-A1-IH	T-20x N4 → product header → P-201A/B	settled wet oil	6" / 8" hdr	120 m³/h	1.0 m/s	HV-2k5 per tank (car-seal line-up)	PS02 P1
301	6"-P-2140-A1-IH	P-201A/B → V-301 nozzle N-F (submerged fill)	wet oil ≥ 70 °C	6" insulated	≤ 60 → 120 m³/h	0.9 → 1.8 m/s	pump HV + NRV; HV-324 at N-F	PS02 P1; PS03 D36
302	12"-P-3302-A1-IH	V-301 cone tip → P-301A/B/C	process oil	12" (8" br.)	434.3 m³/h	1.67 m/s	suction HV + duplex strainer per pump	PS03 P18
303/304	8"-P-3303 / 3304-A1-IH	P-301A/B/C → E-301 → riser → HV-303 → N-R	process oil	8"	434.3 m³/h	3.74 m/s	NRV per pump; HV-301 N.O.; HV-303 2.0 bar(a)	PS03 P18
306	DN450 STD (vapour)	V-301 top → E-302 (one inverted U)	water vapour + NC	DN450	6.53 t/h	28.9 m/s	none (no valve in the vapour line)	PS03 P18
307/313	6"-P-3307 [A] / 3"-W-3313	E-302 → D-301A/B → TK-901 (after the vacuum break)	condensate	6" / 3"	batch 12 min	—	HV-311…315 per receiver	PS03 P18
311	3"-VT-3311 [A]	VP-301 → D-302 → atmosphere	air + water vapour	3" [A]	≈ 100 kg/h	—	none (no valve on the vent)	PS03 P18
DSL	3"-DS-8101 / 2"-DS-8102-A1 [A]	truck → TK-801 → P-802 → TK-301	diesel / solar	3" / 2" [A]	30 / 10 m³/h	≈ 1.3 m/s	HV-811…813; I-803, I-303 trips	G-4; Sheet 13
314	3"-DS-3314-A1	TK-301 → P-302 → V-301 nozzle N-D	diesel / solar	3"	25 m³/h	1.46 m/s	HV-307 N.C. + NRV	PS03 P18
315	8"-P-3315-A1-IH	P-301A/B → TK-601/602 (end of batch)	blended product	8"	≈ 400 m³/h	≈ 3.4 m/s	HV-301 closed; HV-302 (DBB) open; tank HV	PS03 P18
601	8"-PR-6105 / 6"-PR-6108-A1	TK-601/602 → P-601A/B → battery limit (kiln)	fuel product	8" / 6"	≈ 50 m³/h [A]	0.75 m/s (6")	HV-616 at BL + FQI-607	G-1 (duty HOLD)
320/321	8"-HO-3320 / 3321-B1-IH75	US-03 hot oil ↔ E-301 shell side	thermal oil 200 / 170 °C	8"	249 m³/h	2.1 m/s	HV-304, HV-306 N.O.; HV-305 bypass	PS03 P18
CW	12"-CWS-8200 / 10"-CWS-8201, CWR-8202	CT-801 → P-801A/B → users → CT-801	cooling water 32 / 47 °C	12" / 10"	≈ 260 m³/h	≈ 1.4 m/s (10")	P-801 HV + NRV; I-801, I-802	G-5; Sheet 12
330/331	8"-CWS-3330 / CWR-3331-C1	CW header ↔ E-302 tubes	cooling water	8"	238 m³/h	2.0 m/s	HV-325 / HV-326	Sheet 09
901	8"-W-9100 [A] / 3"-W-9102-A1	effluent header (gravity) → TK-901 → P-901A/B → sewer	effluent	8" grav. / 3"	≈ 205 t/day; 30 m³/h	—	pump HV + NRV; HV-904 (sample first)	G-2; Sheet 14
skim	2"-P-9101-A1 [A]	TK-901 skimmer → P-902 → T-101 C1	skimmed oil	2"	2 m³/h	—	HV-902	Sheet 14

KEY FIGURES — SHEET 2

Item	Value
Metallic items	13 tags, 19 units
Total steel (empty)	≈ 146 t
Heaviest lift	V-301 ≈ 46 t
Largest diameter	T-201…T-204 7.0 m
Tallest	V-301 10.82 m tip → crown
Thickest wall	V-301 25 mm (FV)
Max operating load	TK-601/602 ≈ 213 t
Max hydrotest load	TK-601/602 ≈ 280 t
Transport	D > 4.5 m: site fabrication

NOTES

1. Dimensions: GEN-EL-001 Rev P2 (Metallic Vessels Dimensional Register), traced to each DB.
2. Thickness calculated: V-301 (FV, DB 10.4), E-302 (DB 10), TK-601/602 (API 650 one-foot 4.65 → 6 mm). All others are assumptions [A] for the weight estimate only.
3. Empty weights include +10 % for nozzles, manways and clips; V-301 includes trays T-1…T-6, flash box and demister.
4. Operating = empty + liquid at design fill; hydrotest = empty + water full (governs foundations).
5. Weights exclude legs / skirts, foundations, insulation, ladders and platforms (+10-20 % typical).
6. Rev P4 (19-Oct-2026): all tags and line numbers per GEN-PID-001 Rev P1; PS-01 per DB Rev.2 (HOP-101 collar, weir 1.40 m, U-link, P-101 → TK-901 / C2, P-102 ≥ 40 m³/h); line 103 is 3" (P&ID); 601 rated 50 m³/h [A] (Sheet 11); CT-801 header added; TK-901, TK-302 and civil items listed.
7. Valves: plant MANUAL; gate = isolation, globe = throttling, PSV on every PD pump; trips per P&ID Sheet 02.
8. Velocities: ID 8" 202.7, 10" 254.5, 6" 154.1, 4" 102.3, 3" 77.9 mm (ASME B36.10M Sch40).

SEVERO WM OIL PROJECT

Overall PFD

Sheet 2 — Vessel / equipment and line schedule

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PFD-001

Class CL — Client Deliverable

Rev: P4 Status: IFR

Author: Eng. Mohamed Tarek
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Ref.: GEN-PID-001 P1; GEN-EL-001 P2; PS01-DB Rev.2

Date: 19-Oct-2026

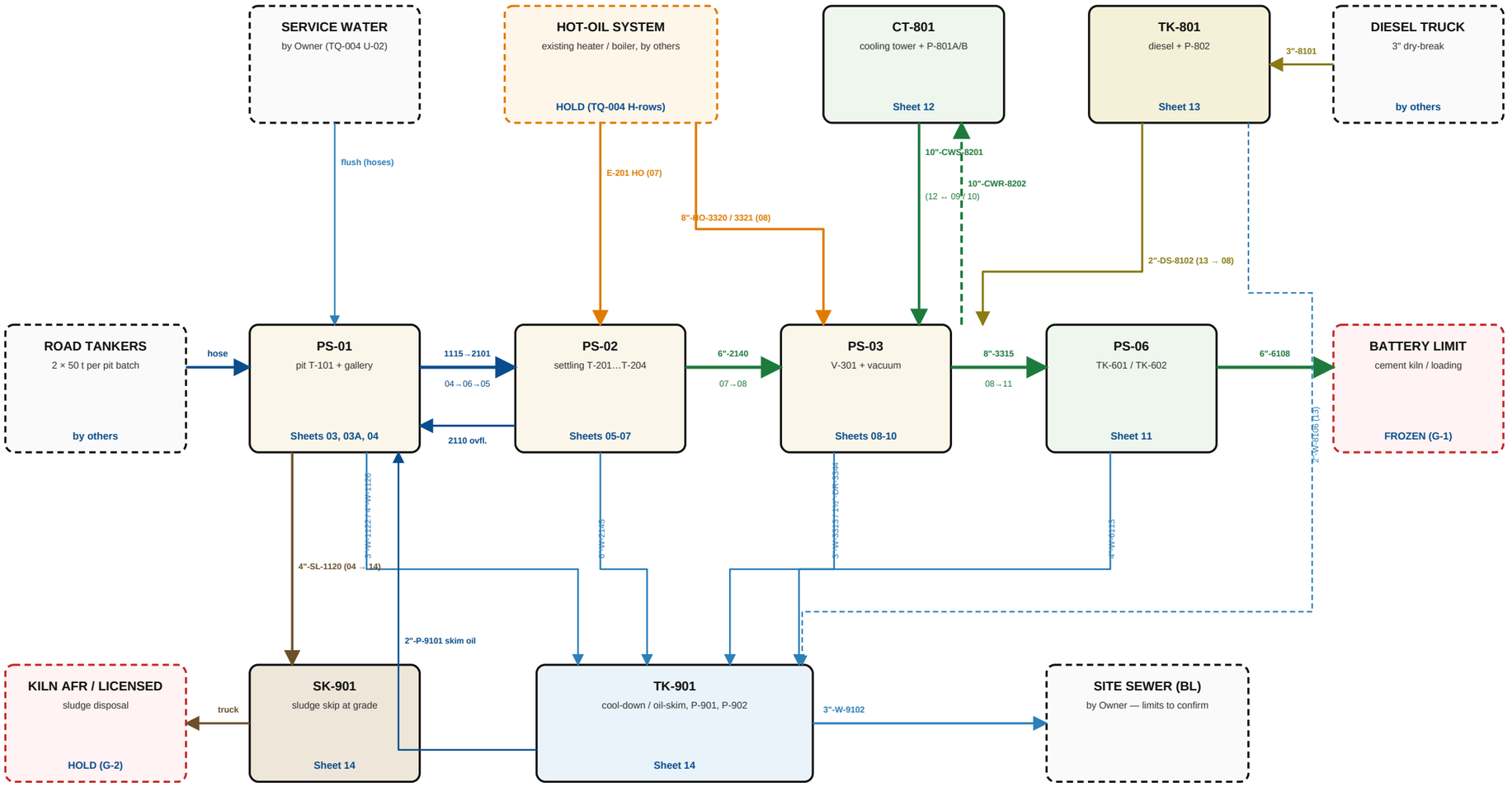
Sheet 2 of 2

A3 — NTS (schematic)

Supersedes Rev P3 (17-Oct-2026)

GENERAL — INTERCONNECTION DIAGRAM: UNITS, UTILITIES, OFFSITES AND INTER-UNIT LINES

Block diagram of the whole plant. Each block = the P&ID sheets shown inside it; each arrow = an inter-unit line (number, size, from/to sheet). Details on the unit sheets; table of inter-unit lines below.



PLANT SUMMARY (design case)

Item	Value
Product (dry-oil basis)	200 t/day out of PS-03 (3 batches)
Raw feed	≈ 407 t/day at 50 % water (8.1 tankers)
Water removed	PS-01 ≈ 82 · PS-02 ≈ 56 · PS-03 ≈ 67 t/day
Product to PS-06	≈ 216.5 t/day (incl. 7 % diesel [A])
Heating	existing hot-oil system (HOLD data)
Cooling	own CT-801, ≈ 4.5 MW [A]
Operation	manual; automatic trips for equipment
	protection only (see each sheet)

ALL TRIPS / INTERLOCKS IN THE PLANT

Tag	Sheet	Function
I-101	03/04	LSLL-103 → stop P-103A/B
I-102	04	LSH-105 → start P-105
I-201	07	PSLL-273 → stop P-202A/B
I-202	07	PSLL-283 → stop P-201A/B
I-301	08	LSLL-345 → stop P-301A/B/C
I-302	10	CH-301 pump stopped → stop VP-301
I-303	08/13	LSHH-347 (TK-301) → stop P-802
I-601	11	PSLL-606 → stop P-601A/B
I-801	12	LSLL-801 → stop P-801A/B
I-802	12	VSHH-802 → stop CT fan
I-803	13	LSLL-823 → stop P-802
I-901	14	LSLL-902 → stop P-901A/B

NOTES

1. Arrows show the normal flow direction; dashed = return.
2. Battery limits (BL): beyond them FROZEN / by Owner.
3. All alarms (LAH, TAH, XA) are listed on the unit sheets.
4. Service water also feeds CT-801 make-up (2"-W-8203, Sh.12), PS-03 seal-loop make-up (1"-W-3345, Sh.10) and hose flushing.

INTER-UNIT LINES (continuations between sheets)

Line	Service	From (sheet)	To (sheet)	Remarks
8"-P-1115-A1 → 8"-P-2101-A1	cold wet feed, 80 m³/h	P-103A/B (04)	ST-201 (06) → N1 header (05)	X-201 quill on 1115; not insulated
8"-P-2110-A1	PS-02 overflow, gravity	T-201…204 N10 (05)	PS-01 pit C1 (03)	no valve
6"-P-2140-A1-IH	wet charge ≥ 70 °C	P-201A/B (07)	V-301 N-F (08)	≤ 60 m³/h until N-F covered
8"-P-3315-A1-IH	product ≈ 90 °C	HV-302 DBB (08)	TK-601 / 602 (11)	by P-301, vessel at atmosphere
6"-PR-6108-A1	product send-out	P-601A/B (11)	battery limit	FROZEN (G-1), FQI-607 at BL
2"-DS-8102-A1 [A]	diesel	P-802 (13)	TK-301 (08)	I-303 overfill trip
10"-CWS-8201 / CWR-8202-C1 [A]	cooling water ≈ 260 m³/h	CT-801 (12)	E-302, D-301, E-303 (09); CH-301 (10)	32 → 47 °C
8"-HO-3320 / 3321-B1; E-201 HO	hot oil	existing system	E-301 (08); E-201 (07)	HOLD (TQ-004 H-rows)
3"-W-1122 · 4"-W-1126 · 6"-W-2145 · 3"-W-3313	effluent water	P-102/P-101 (04) · T-20x (05) · D-301 (09)	TK-901 (14)	6"-W-2145 hot 80 °C, liquid seal
1½"-DR-3344 · 4"-W-6113 · 2"-W-8106	oily drains	TK-302 (10) · PS-06 (11) · TK-801 (13)	TK-901 (14)	gravity
2"-P-9101-A1 [A]	skimmed oil	P-902 (14)	PS-01 pit C1 (03)	re-processed
4"-SL-1120-A1 [A]	sludge	P-101A/B (04)	SK-901 (14)	disposal HOLD (G-2)
3"-W-9102-A1	treated effluent	P-901A/B (14)	site sewer (BL)	T ≤ 43 °C, limits to confirm
DN450-VA-3306-V1	vapour 6.5 t/h	V-301 crown (08)	E-302 (09)	full-vacuum design
2½"-VA-3309 · 2½"-P-3341	vacuum gas / VP discharge	E-303 (09) · VP-301 (09)	VP-301 (10) · D-302 (10)	

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 02 — Interconnection diagram

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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Ref.: Sheets 03-14; GEN-PFD-001 Rev P4

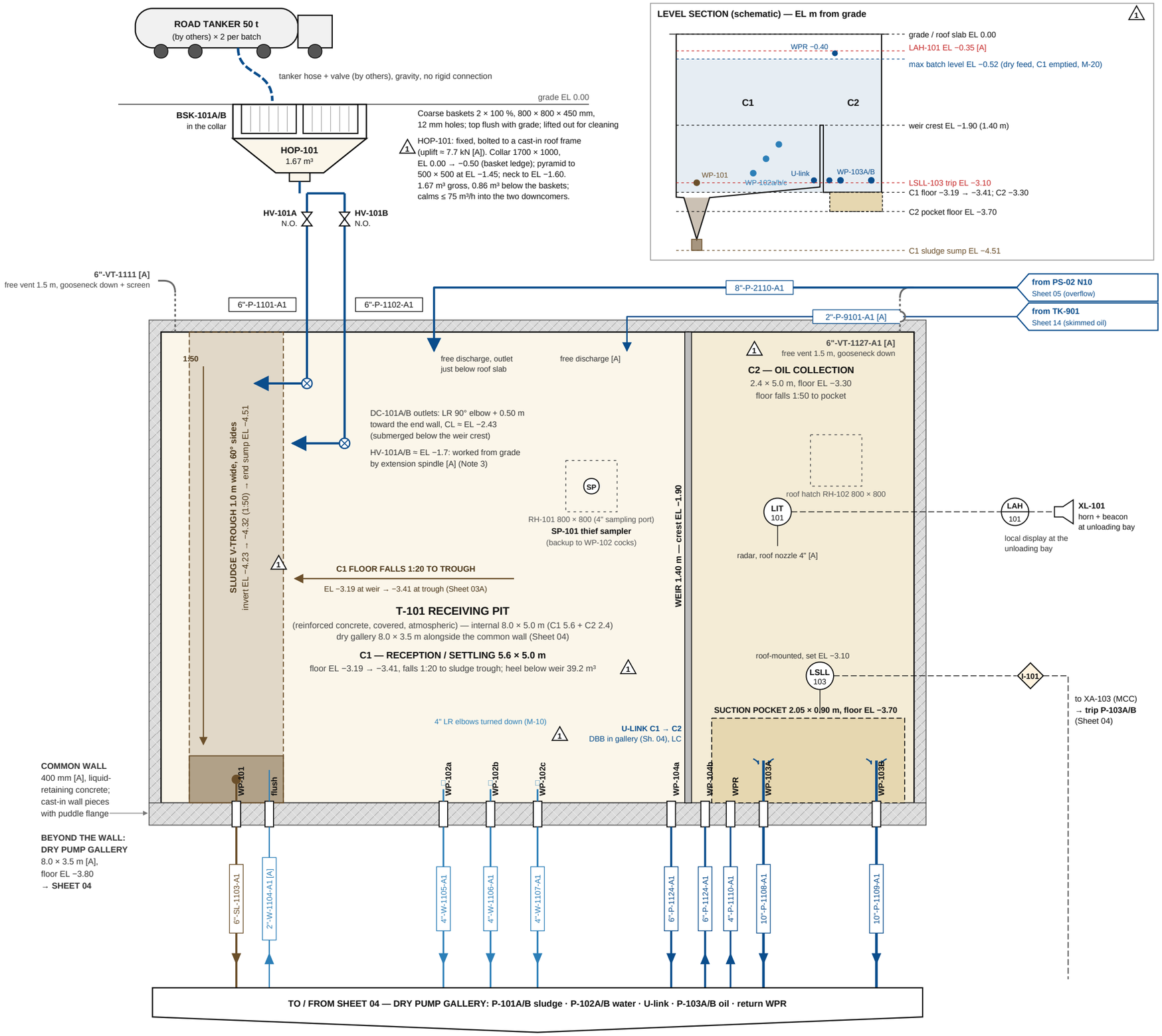
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Sheet 02 of 15

A3 — NTS (schematic)

PS-01 — TANKER UNLOADING AND RECEIVING PIT T-101 (PLAN VIEW + LEVEL SECTION)

P&ID, NOT TO SCALE. Basis: PS01-DB-001 Rev.2 / PS01-LAY-001 Rev.2 + owner review 19-Oct (M-04…M-20). Wall pieces continue on Sheet 04; floor falls, sections and pit details on Sheet 03A.



KEY DATA — SHEET 03

Item	Value
Pit T-101 internal	8.0 × 5.0 × 3.30 m (C1 5.6 / C2 2.4)
Batch	100 t = 2 tankers × 50 t (111.1 m³)
Max batch level / freeboard	EL −0.52 / 0.52 m (C1 emptied, M-17/M-20)
Weir / C1 heel	1.40 m, crest EL −1.90 / 39.2 m³
Unloading design flow	≤ 75 m³/h, gravity
Water draw-off during unloading	P-102 ≥ 40 m³/h from WP-102a (M-12)
Feed water content	26 vol% (lab) … 50 % (design)
Feed temperature	ambient, 10 … 40 °C [A]
Emulsion viscosity	2.48 Pa·s @ 20 °C; 6.0 @ 10 °C
Flash point	130 °C (Owner value, confirm)
Design	atmospheric, ambient; class A1

TRIP / INTERLOCK AND ALARM SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-101	LSLL-103 (C2)	EL −3.10	Stop P-103A/B	Manual, LCS
LAH-101	LIT-101 (C2)	EL −0.35 [A]	Horn XL-101	Self

MOTOR DATA

Tag	kW	Starter	Control	Trip by
—			No motors (pumps on Sheet 04)	

NOTES

1. Plan view, NTS. EL in m from finished grade 0.00. Wall pieces are cast into the common wall; lines continue on Sheet 04.
2. Unloading: 2 tankers × 50 t, gravity through the tanker hose (by others) into BSK-101A/B. P-102 (≥ 40 m³/h) RUNS from WP-102a for the whole unloading; P-103A/B do NOT run during unloading.
3. HV-101A/B (6" gate, N.O.) sit below the hopper (≈ EL −1.7): worked from grade by extension spindle [A]. Close one downcomer for cleaning; the other carries 75 m³/h alone.
4. Unloading rules (M-12): tanker 2 starts only when the WP-102b cock shows oil. Water at the WP-102c cock → stop unloading (tanker valve) and start the standby P-102.
5. End of batch (M-13): settle → P-101 sludge → skip, then bottom water → TK-901 → P-102 to oil (throttle HV-110) → rag layer to C2 (WPR) → U-link open (bleed cock = oil) → P-103 to LSLL-103 → P-101 pumps the C1 residual (≈ 7.8 m³ below EL −3.10) to C2 via HV-120 (M-20) → P-103 to LSLL-103 → U-link LC.
Sludge MUST be removed every batch. Cycle ≈ 5.9 h.
C1 MUST be emptied (U-link + P-101 residual) before unloading (M-17): with the C1 heel full a 100 t batch does not fit the pit.
6. LSLL-103 = the only automatic trip in PS-01 (dry-run protection of P-103A/B). Below the weir C2 falls 6.7 m/h at 80 m³/h.
7. LIT-101 / LAH-101 / XL-101: alarm only, no trip (approved) — tells the operator to stop unloading before the freeboard is used.
8. Floor and trough falls, sections, U-link, hatches, two vents and HOP-101 geometry: see Sheet 03A. Flush 2" (P-102 discharge, Sh. 04) into the sludge dip pipe and into the U-link C1 leg.
10. Renumbered: BSK-1/2 → BSK-101A/B, HOP-1 → HOP-101, DC-1/2 → DC-101A/B (lines 1101/1102), SP-1 → SP-101.

WALL-PIECE SCHEDULE (common wall, cast-in)

WP	Size	CL EL	Inside the pit	Line
WP-101	6"	−3.10	dip pipe to sump, lip −4.41	6"-SL-1103
flush	2"	−3.10 [A]	into dip pipe / sump	2"-W-1104
WP-102a	4"	−2.90	4" LR elbow down, C1	4"-W-1105
WP-102b	4"	−2.60	4" LR elbow down, C1	4"-W-1106
WP-102c	4"	−2.30	4" LR elbow down, C1	4"-W-1107
WP-104a	6"	−3.05 [A]	U-link inlet, C1 at weir	6"-P-1124
WP-104b	6"	−3.05 [A]	U-link outlet, C2 at weir	6"-P-1124
WPR	4"	−0.40	return outlet, C2	4"-P-1110
WP-103A	10"	−3.05	14" bell down, lip −3.55	10"-P-1108
WP-103B	10"	−3.05	14" bell down, lip −3.55	10"-P-1109

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 03 — PS-01 Tanker unloading and receiving pit

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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Ref.: PS01-DB-001 Rev.2; LAY-001 Rev.2; Sh. 01, 03A, 04

Date: 19-Oct-2026

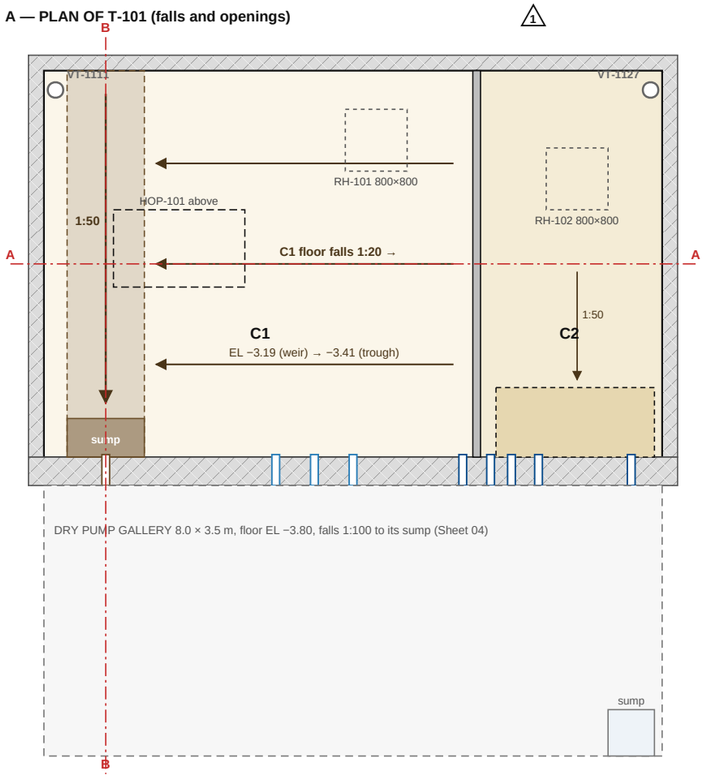
Sheet 03 of 15

A3 — NTS (schematic)

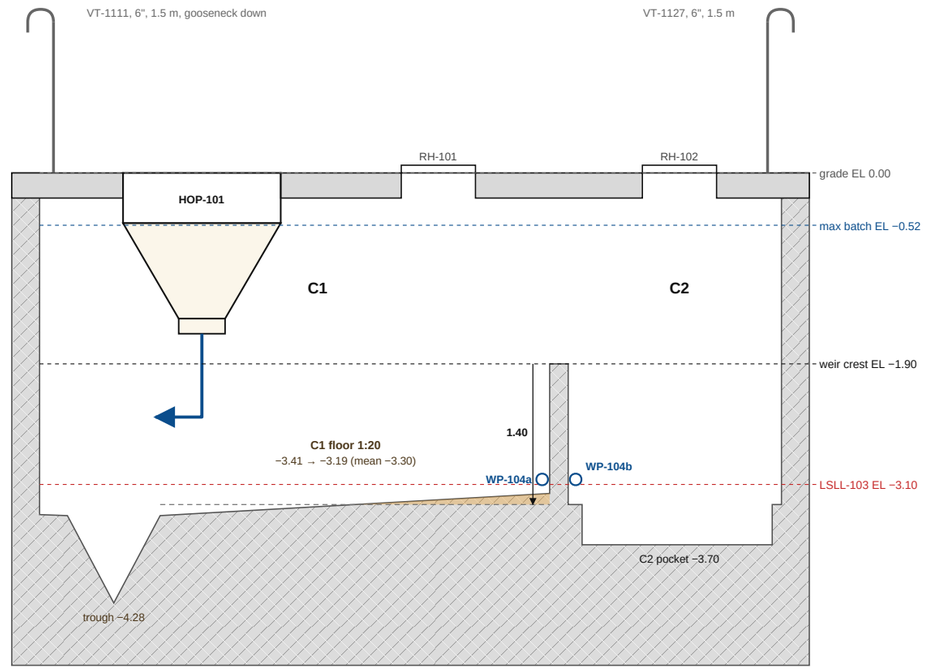
PS-01 — RECEIVING PIT T-101 DETAILS: FLOOR FALLS, SECTIONS, U-LINK, HOPPER, HATCHES, VENTS

Detail sheet to Sheet 03 (new in Rev P1, owner review 19-Oct: M-04, M-07, M-08, M-11 … M-16). Schematic, NOT TO SCALE — elevations govern; vertical scale exaggerated in the sections.

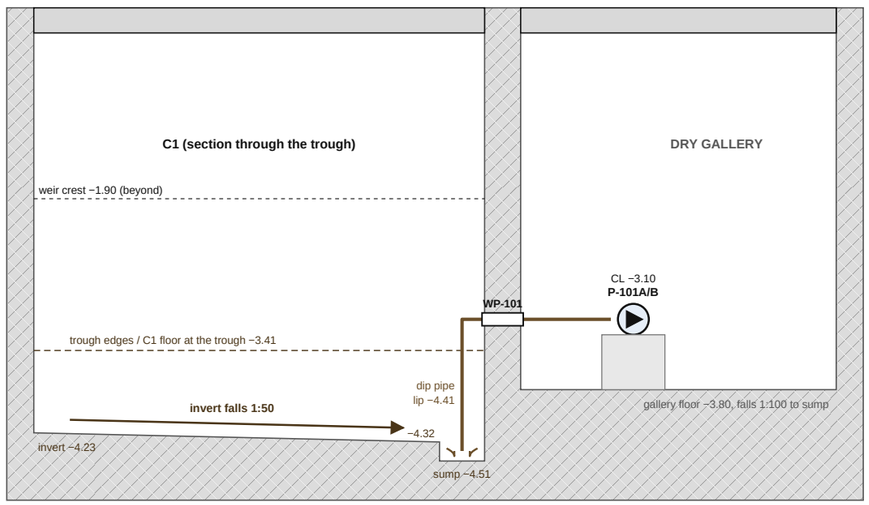
A — PLAN OF T-101 (falls and openings) 1



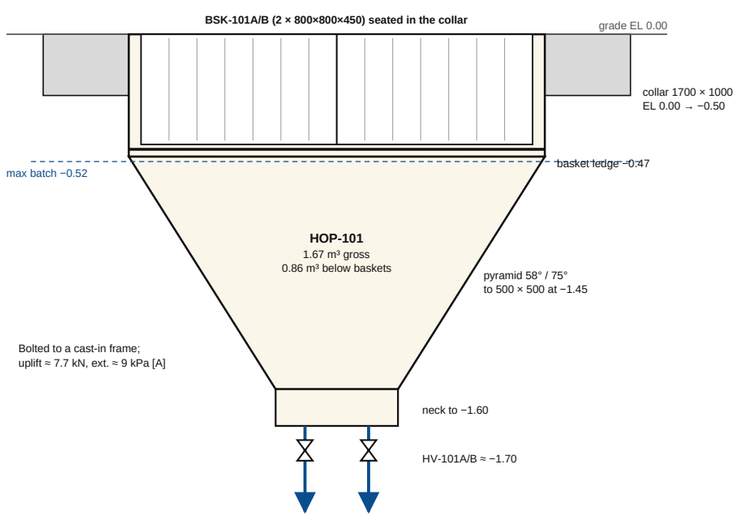
SECTION A-A — along the pit length (looking at the common wall) 1



SECTION B-B — along the sludge trough (across the pit) into the dry gallery 1



D — HOP-101 SECTION (long side, 1700 mm) 1



KEY ELEVATIONS — T-101 (EL m from grade)

Item	EL
Grade / roof slab top; slab 250 mm [A]	0.00 / soffit −0.25
LAH-101 / max batch level (C1 emptied)	−0.35 [A] / −0.52
HOP-101 collar / pyramid / neck bottom	−0.50 / −1.45 / −1.60
HV-101A/B (extension spindle)	≈ −1.70
Weir crest (1.40 m above mean floor)	−1.90
Downcomer outlets CL	−2.43
WP-102c / b / a (4" elbow down)	−2.30 / −2.60 / −2.90
U-link WP-104a/b CL	−3.05 [A]
WP-101 / P-101A/B CL; LSLL-103	−3.10
C1 floor (weir → trough); C2 floor	−3.19 → −3.41; −3.30
C2 suction pocket / gallery floor	−3.70 / −3.80
Trough invert (far wall → sump end)	−4.23 → −4.32
Dip-pipe lip / end sump floor	−4.41 / −4.51

NOTES

1. Schematic sections; vertical scale exaggerated. Elevations govern.
2. C1 floor: benching on a flat raft, falls 1:20 [A] to the trough, volume-neutral about EL −3.30 (heel and batch level unchanged). Final gradient after the sludge tests (TQ: ASTM D4007 / D473, yield stress); at 1:10 the floor is −3.085 / −3.515.
3. Sludge V-trough 1.0 m wide, 60° sides; invert 1:50 to the end sump 1.0 × 0.5 m. Sludge must be removed every batch (P-101).
4. Weir 1.40 m above the mean floor (1.29 m at its face), crest EL −1.90; C1 heel 39.2 m³; one tanker of free water (24.4 m³) is held even with P-102 down (M-12).
5. U-link WP-104a/b (6", line 1124): DBB HV-115 knife gate + HV-116 gate, both LC, bleed SP-102, 2" flush HV-117 (Sheet 04). Open only after sludge and water are removed; NEVER while unloading. MANDATORY every batch (M-17): C1 emptied before the next unloading; the ≈ 7.8 m³ below EL −3.10 is pumped to C2 by P-101 / HV-120 (M-20).
6. Roof hatches RH-101 / RH-102: 800 × 800 clear (EN 476 ≥ 600), hinged, gasketed, lockable, EN 124 D400 [A]. RH-101 cover with a 4" sampling port for SP-101. Confined-space entry by permit.
7. Vents VT-1111 (C1 end) / VT-1127 (C2 end): 6", vertical through the roof slab in a cast-in sleeve with puddle flange; outlet 1.5 m above grade, gooseneck facing down, SS screen; away from walkways and the tanker bay (≈ 2.2 m if beside a walkway). IIIB: no arrester.
8. HOP-101 fixed: bolted to a cast-in roof frame (uplift, ext. pressure).
9. Wash-down: 2-3 hose points from the P-102 flush header [A]; vacuum truck as backup through RH-101 / RH-102.
10. Dry gallery: floor falls 1:100 to its sump; P-105 auto (Sheet 04).

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 03A — PS-01 Receiving pit details

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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Ref.: Sh. 03, 04; PS01-DB-001 Rev.2; PS01-LAY-001 Rev.2

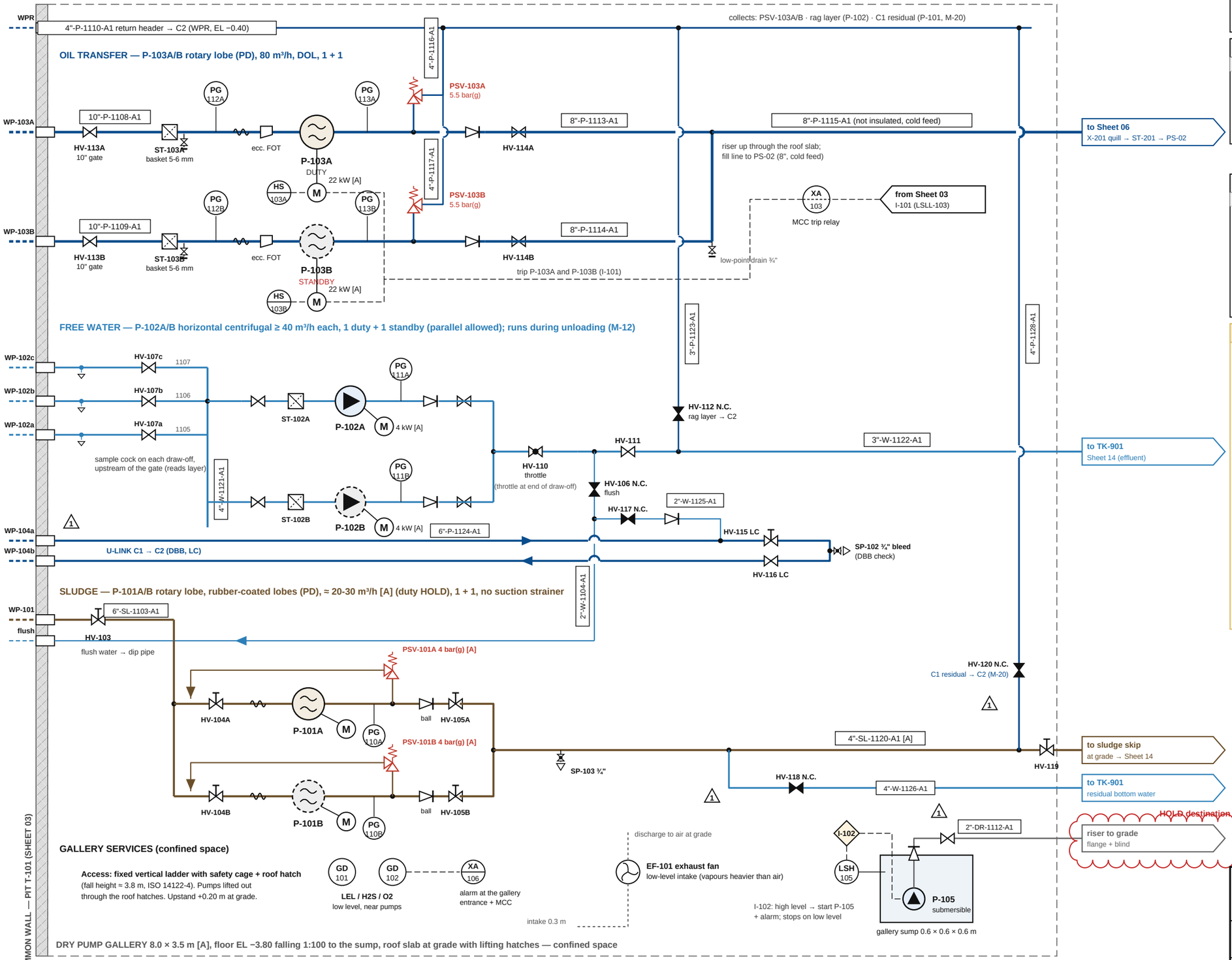
Date: 19-Oct-2026

Sheet 03A of 15

A3 — NTS (schematic)

PS-01 — DRY PUMP GALLERY: OIL TRANSFER P-103A/B, WATER P-102A/B, SLUDGE P-101A/B, GALLERY SERVICES

P&ID, NOT TO SCALE. Basis: PS01-DB-001 Rev.2 §10-12, PS01-LAY-001 Rev.2 + owner review 19-Oct (M-10, M-12, M-13, M-16, M-20). Left = common wall with pit T-101 (Sheet 03). Right = risers through the gallery roof to grade.



TRIP / INTERLOCK AND ALARM SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-101	LSLL-103 (Sh.03)	EL −3.10	Stop P-103A/B	Manual LCS
I-102	LSH-105 sump	high [A]	Start P-105 + alarm	Auto
XA-106	GD-101/102	LEL 10 % [A]	Alarm only	Self
		H2S 10 ppm [A]		
		O2 < 19.5 %		

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-103A/B	22 [A]	DOL *	LCS HS-103A/B	I-101
P-102A/B	4 [A]	DOL	LCS HS-102A/B	—
P-101A/B	7.5 [A]	DOL *	LCS HS-101A/B	—
P-105	0.75 [A]	DOL	auto I-102 + LCS	—
EF-101	0.75 [A]	DOL	LCS at entrance	—

** PD pump: direct-on-line, NOT star-delta (full torque on cold oil).*
Supply 400 V / 3 ph / 50 Hz [A]. Motor kW per API 676 at PSV set.

KEY DATA — SHEET 04

Item	Value
P-103 duty	80 m³/h rated, TDH ≈ 3.8 bar winter
P-103 suction	10" per pump, NPIPa ≈ 0.74 bar(a)
P-103 discharge	8" per pump → 8" common riser
PSV-103A/B	5.5 bar(g), outlet → return header
P-102 duty	≥ 40 m³/h each (fixed, M-12)
P-101 duty	HOLD (sludge data, TQ-004)
Fluid	cold wet oil 2.5-6.0 Pa·s; water; sludge
Area classification	unclassified [A] (Note 6)

NOTES

1. Wall pieces WP-101/102/103/WPR and flush: see Sheet 03 schedule.
2. Unloading (M-12): P-102 duty runs from WP-102a for the whole unloading; P-103 never runs while unloading. End of batch (M-13): P-101 sludge → skip, then bottom water → TK-901 (HV-119 shut, HV-118 open, SP-103) → P-102 to oil → rag (HV-112 to WPR) → U-link → P-103 to LSLL-103 → P-101 C1 residual → C2 (HV-120, HV-118/119 shut, M-20) → P-103 to LSLL-103.
3. P-102: open the lowest draw-off that shows water at its sample cock; throttle with HV-110 at the end of the draw-off.
4. Every PD pump has its own PSV upstream of any valve: PSV-103A/B to the return header (C2), PSV-101A/B to its own suction.
5. HV-106 / HV-112 / HV-117 / HV-118 / HV-120 N.C.: flushing, rag, bottom water, C1 residual. U-link HV-115 / HV-116 LC: open only after SP-102 shows oil; NEVER while unloading. U-link emptying of C1 is MANDATORY at the end of every batch (M-17).
6. Flash point 130 °C (Owner value) > handling temperature: gallery unclassified [A]. If lighter deliveries are possible → Zone 2.
7. Gallery entry: confined-space permit; EF-101 running ≥ 15 min before entry; GD-101/102 healthy.
8. ¾" drains at strainer bodies and low points; vents at high points.
9. P-105 discharge (M-16): riser to grade, flange + blind; destination HOLD (outside, or the PS-02 tundish / effluent drain → TK-901). Gallery floor falls 1:100 to the sump; P-105 auto by I-102.

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 04 — PS-01 Dry pump gallery

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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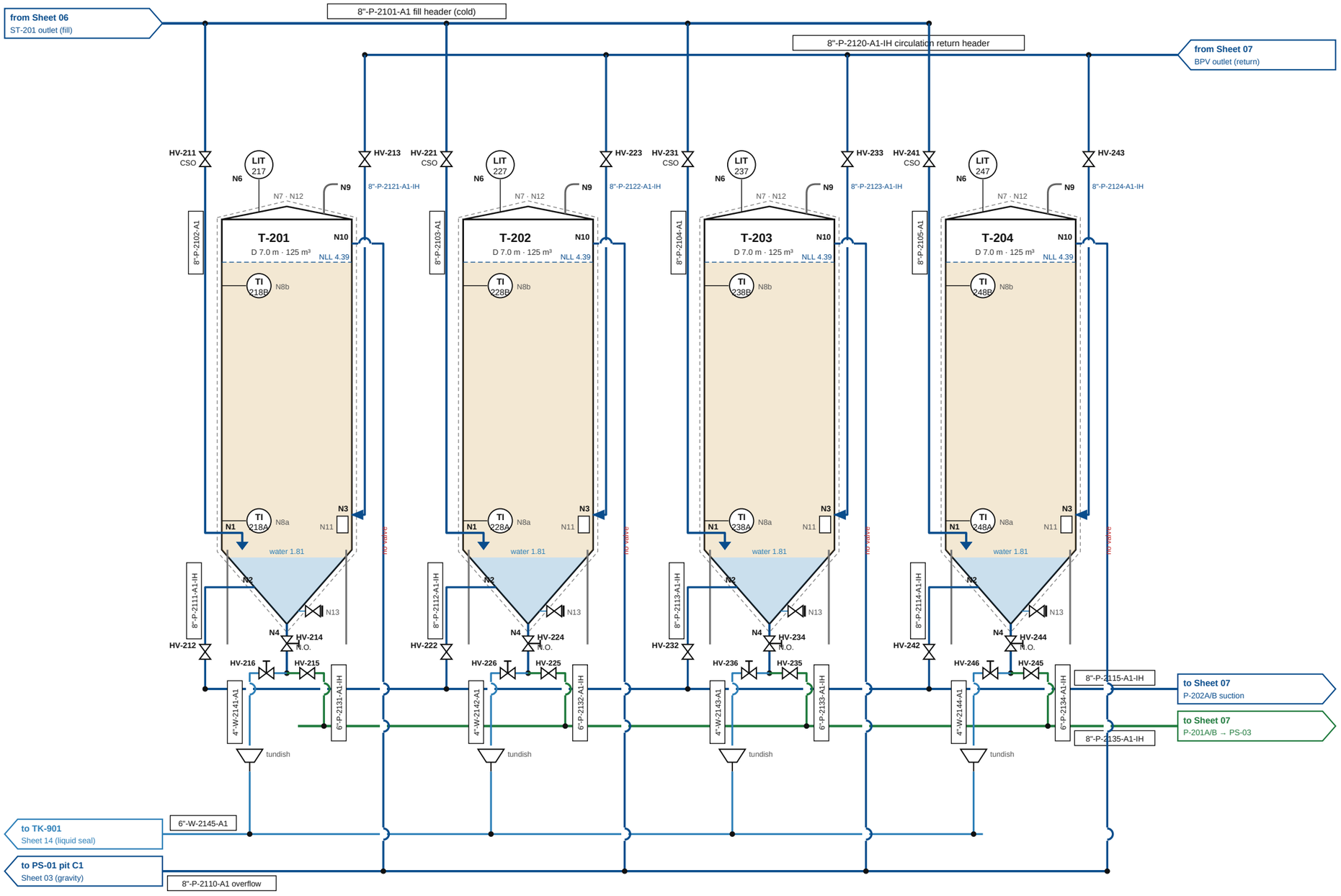
Date: 19-Oct-2026

Sheet 04 of 15

A3 — NTS (schematic)

PS-02 — HEATED SETTLING TANKS T-201 … T-204 (4 × 125 m³), NOZZLES AND TANK HEADERS

P&ID, NOT TO SCALE. Basis: PS02-DB-001 Rev P1 (D-Q … D-W, D-AD). Fill header from Sheet 06; heater loop and transfer pumps on Sheet 07. Elevations EL in m from the cone tip CT (= +1.50 m above grade).



Dashed outline around each tank = 50 mm rock wool + Al cladding (shell, cone, roof). Valve tags HV-2kn: k = tank 1…4, n = 1 fill · 2 suction · 3 return · 4 bottom isolation · 5 product · 6 water drain.

KEY DATA — T-201 … T-204 (each)

Item	Value
Geometry	D 7.0 m, cone 30°, cone-roof, elevated
Max working volume / freeboard	125 m³ / 0.5 m [A]
Charge (design, Poor case)	108.33 t wet = one PS-03 charge
Levels (EL from CT)	water 1.81 · NLL 4.39 · max 4.60
Shell / cone-shell junction / top	EL 2.02 → 5.10
Heating	20 → 80 °C, ≈ 0.91 MW, ≈ 5.6 h
Settling	≤ 16 h quiescent, d ≥ 271 µm
Design P / T	atm. +0.5 / −0.25 kPa(g) [A]; 90 °C
Code / thickness	API 620 [A]; shell 6, cone 8, roof 6 mm
Weights empty / oper. / test	8.6 / 117 / 153 t
Support	8 legs on ring girder at junction
Insulation	50 mm rock wool + Al cladding

NOZZLE SCHEDULE (each tank, D-V / D-AD)

N	Service	Size	EL (CT)
N1	Fill, internal downward diffuser ≤ 1 m/s	8"	2.30
N2	Circulation suction, vortex breaker	8"	1.00 cone
N3	Circulation return, tangential	8"	2.60
N4	Bottom outlet at cone tip → drain / product	6"	0.00
N6	Radar level LIT-2k7, local display	4"	roof
N7	Gauge hatch (dip + sampling)	8"	roof
N8a/b	Dial thermometers TI-2k8A/B	1½"	2.50 / 4.00
N9	Free vent, gooseneck + screen	6"	roof
N10	Overflow by gravity → PS-01 (no valve)	8"	4.70
N11 / N12	Shell / roof manway	24"	2.70 / roof
N13	Cone flush, gate + blind (hose)	2"	≈ 0.35
N14a/b	Spare, blinded	3"	shell / roof

TRIP / INTERLOCK AND MOTOR DATA

Trips	none on this sheet (P-202 / P-201 trips: Sheet 07)
Motors	none on this sheet

NOTES

1. N9 vent 6" checked to API 2000: in-breathing ≈ 145 Nm³/h (P-201 120 m³/h + thermal ≈ 25), out-breathing ≈ 85 Nm³/h (fill 80 m³/h + thermal) → ΔP < 0.05 kPa: adequate (O-2 closed).
2. N10 overflow and N9 vent: NO valve, ever.
3. Water draw-off: 4" ≈ 140 m³/h → 18.6 m³ in ≈ 8 min; hot water at the open tundish: splash cover + PPE; outlet closed pipe with liquid seal to TK-901.
4. N4 bottom isolation HV-2k4 (knife-gate, N.O.) lets the drain and product valves be maintained without emptying the tank.

LINE-UP BY STEP (one tank at a time; others isolated) — valves of tank k

Step	HV-2k1 fill	HV-2k2 suct.	HV-2k3 return	HV-2k4 bottom	HV-2k5 product	HV-2k6 drain	Pumps running	End point
1 Fill + heat	OPEN	OPEN	OPEN	open	closed	closed	P-103 (Sh.04), P-202 + E-201	level NLL and 80 °C (TI-2k8A/B)
2 Settle	closed	closed	closed	open	closed	closed	none (quiescent)	≤ 16 h
3 Water draw-off	closed	closed	closed	open	closed	OPEN (throttle by eye)	none (gravity)	oil seen at the tundish
4 Transfer to PS-03	closed	closed	closed	open	OPEN	closed	P-201 (Sh.07)	LI low; one tank = one PS-03 charge

Heater loop serves ONE tank at a time (N2 → P-202 → E-201 → BPV → N3, Sheet 07). Stop sequence: hot oil off first, P-202 runs ≥ 10-15 min, then stop.
Wrong line-up protection = operating procedure + colour tags + car-seals (CSO on the fill valves of the tank in service), written in the operating manual (D-J).

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 05 — PS-02 Settling tanks T-201 … T-204

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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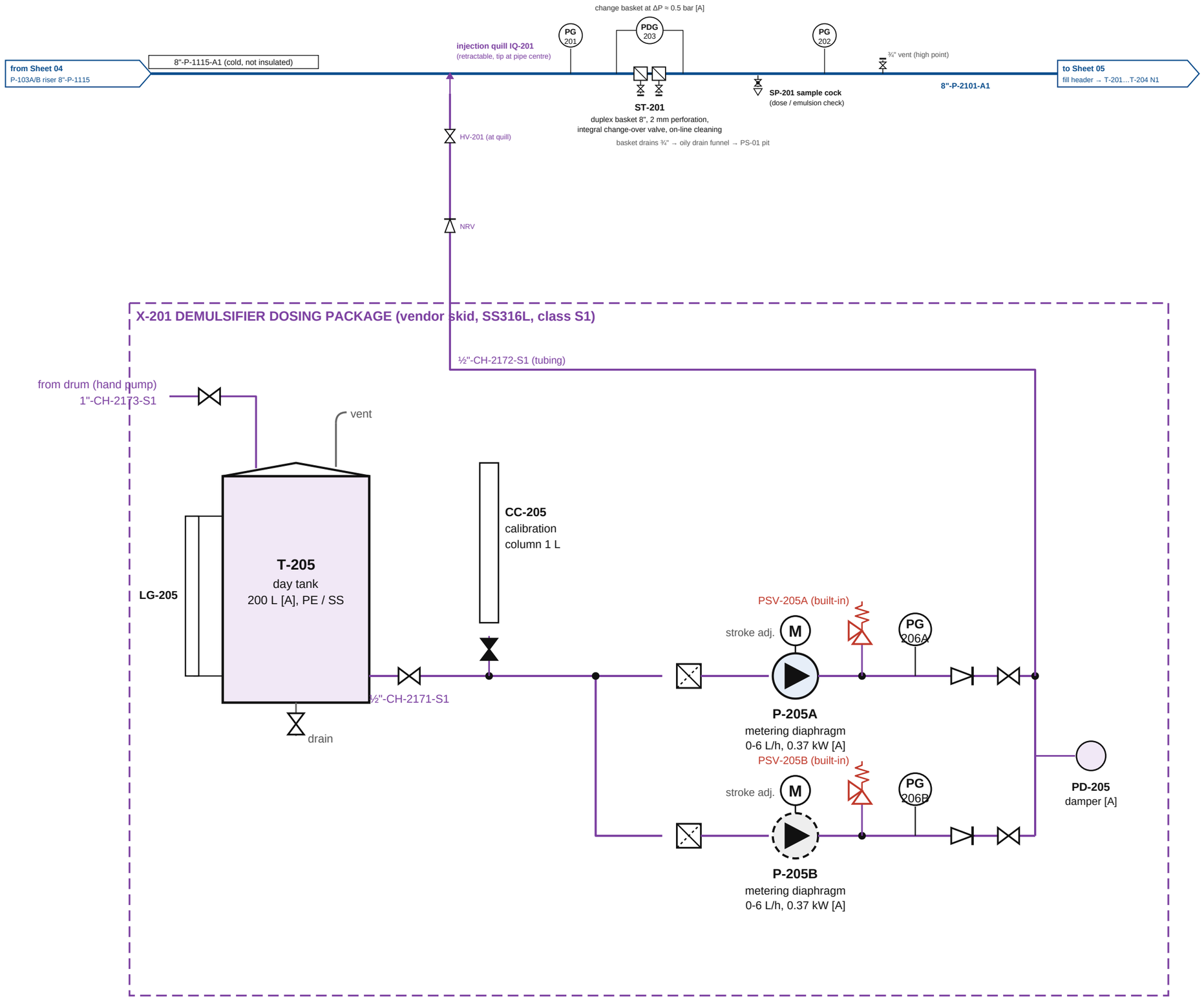
Date: 19-Oct-2026

Sheet 05 of 15

A3 — NTS (schematic)

PS-02 — FILL LINE FROM PS-01: DEMULSIFIER DOSING PACKAGE X-201 AND DUPLEX STRAINER ST-201

P&ID, NOT TO SCALE. Basis: PS01-DB-001 Rev.2 §11; PS02-DB-001 Rev P1 (D-AC, D-AD); O-3 closed on this sheet. Cold wet feed, 80 m³/h rated (P-103A/B fixed speed).



KEY DATA — SHEET 06

Item	Value
Feed flow	80 m³/h rated (≈ 72 t/h), cold
Viscosity (winter / 20 °C)	6.0 / 2.48 Pa·s (emulsion)
ST-201	8" duplex, 2 mm, ΔP clean ≈ 0.3 bar
Demulsifier	polymeric EO/PO resin (backup PEI) [A]
Dose	10-30 ppm, up to 50 ppm (Jar test)
Dosing rate	0.7-3.6 kg/h ≈ 0.8-3.8 L/h
Day tank use	≤ 17 L/day at 50 ppm → 200 L ≈ 12 days
Mixing	by P-202 recirculation in the tank
Package material	SS316L tubing / PVDF pump head [A]

TRIP / INTERLOCK SCHEDULE

—	none: dosing started / stopped with P-103 by procedure

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-205A/B	0.37 [A]	DOL	LCS on skid	—

NOTES

1. O-3 CLOSED: X-201 = vendor skid — day tank T-205, calibration column, 2 × metering diaphragm pumps (1 + 1) with built-in relief, damper, NRV + isolation at the quill. Rate set by stroke (fixed-speed P-103 → constant flow → constant ppm).
2. Dosing runs only while P-103 runs (operating procedure).
3. Dose set from the Jar test (TQ-004 L-02); start at 20 ppm.
4. Demulsifier: handle with gloves / goggles; drum store bunded.
5. ST-201: switch baskets on PDG-203 ≈ 0.5 bar [A]; clean baskets drain to the oily drain funnel.
6. Line 8"-P-1115 runs from the P-103 riser (Sheet 04) to ST-201; 8"-P-2101 starts at the ST-201 outlet (Sheet 05 header).
7. No insulation (cold feed, heated later in the tanks).

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 06 — PS-02 Fill line, X-201 dosing, ST-201

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Rev: P1 Status: IFR

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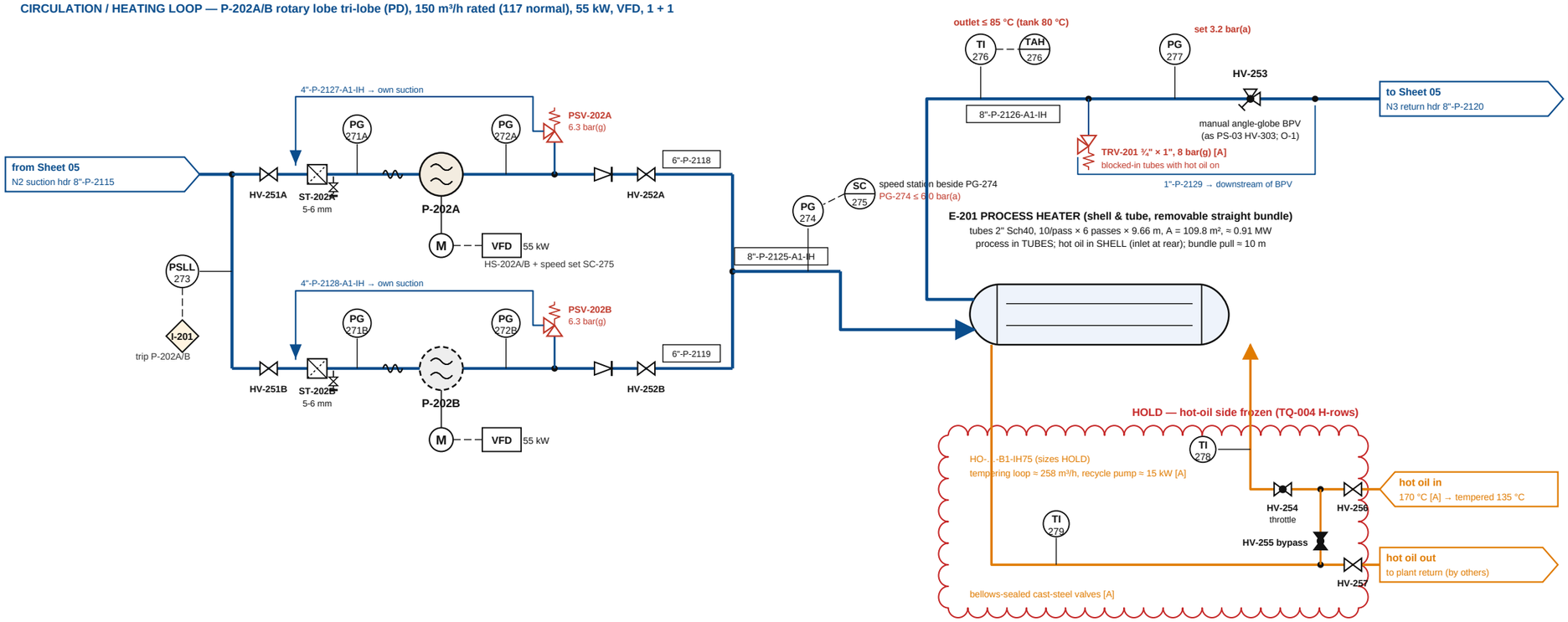
Sheet 06 of 15

A3 — NTS (schematic)

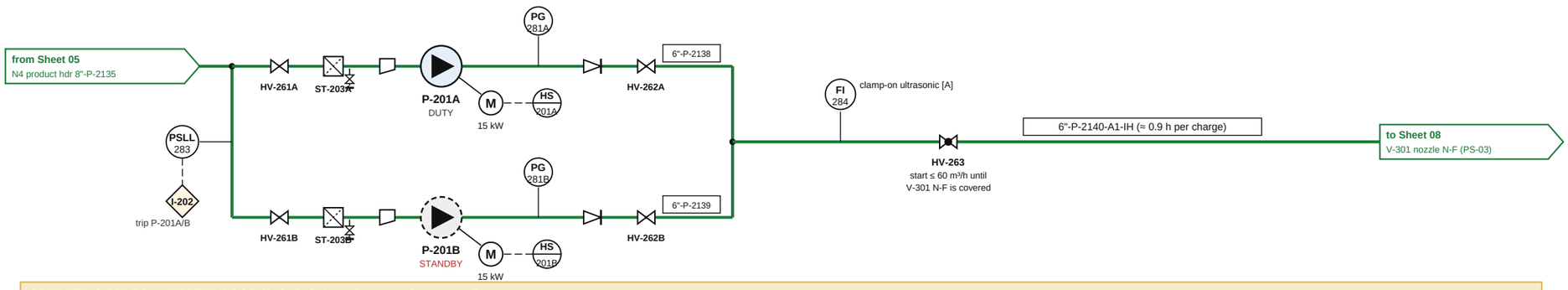
PS-02 — HEATER LOOP E-201 / P-202A/B / BACK-PRESSURE VALVE, AND TRANSFER PUMPS P-201A/B TO PS-03

P&ID, NOT TO SCALE. Basis: PS02-DB-001 Rev P1 (D-E … D-P, D-X). Heater serves ONE tank at a time (line-up on Sheet 05). Hot-oil side FROZEN until plant data (TQ-004 H-rows).

CIRCULATION / HEATING LOOP — P-202A/B rotary lobe tri-lobe (PD), 150 m³/h rated (117 normal), 55 kW, VFD, 1 + 1



TRANSFER TO PS-03 — P-201A/B centrifugal end-suction, SEMI-OPEN impeller, 120 m³/h rated, TDH 15 m, 15 kW, 1 + 1



TRIP / INTERLOCK AND ALARM SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-201	PSLL-273	0.1 bar(g) [A]	Stop P-202A/B	Manual LCS
I-202	PSLL-283	0.1 bar(g) [A]	Stop P-201A/B	Manual LCS
TAH-276	TI-276	85 °C	Alarm at heater	Self

I-201 / I-202: dry-run protection (same rule as P-103 in PS-01).

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-202A/B	55	VFD	LCS + SC-275	I-201
P-201A/B	15	DOL	LCS HS-201A/B	I-202
Tempering P	≈ 15 [A]	HOLD	HOLD	—

KEY DATA — SHEET 07

Item	Value
Duty	≈ 0.91 MW; tank 20 → 80 °C in ≈ 5.6 h
E-201 tube side	inlet ≤ 6.0 bar(a), outlet 3.2 bar(a)
Tube ΔP cap / design	2.8 bar / 8 bar(g), 200 °C [A]
Hot oil	plant RETURN 170 °C [A] → 135 °C in
P-202 PSV	6.3 bar(g) → own suction
P-201	120 m³/h, TDH 15 m, SiC seal, no PSV
Lines	class A1, IH 50 mm; E-201 + HO IH75

NOTES

1. O-1 closed: BPV = manual angle globe HV-253 + PG-277.
2. TRV-201 protects the tube side against blocked-in liquid heated by the hot oil (thermal expansion).
3. Pulsation snubbers on all PGs of the P-202 loop.
4. Hot-oil side: tags, sizes and tempering loop HOLD until TQ-004 H-rows; boiler supply/return untouched (cascade).

OPERATING RULES — HEATER LOOP (D-G, D-G-1; to the operating manual)

1. P-202 runs at normal speed whenever hot oil is on. VFD is for the cold-start ramp only (keep PG-274 ≤ 6.0 bar(a)), NOT for temperature control.
2. Temperature is controlled on the HOT-OIL side only: throttle HV-254 / open bypass HV-255 so that TI-276 ≤ 85 °C (TAH-276 alarm).
3. Stop sequence: hot oil OFF first (close HV-254, open HV-255) → P-202 keeps running ≥ 10-15 min → then stop P-202.
4. Never stop P-202 with hot oil flowing through E-201. Never start P-202 on a tank whose N2 is not covered (≈ 7 m³ minimum).
5. HV-253 (BPV) is set by hand at commissioning for 3.2 bar(a) on PG-277 at normal flow → ≥ 10 °C sub-cooling margin inside E-201, then left fixed.
6. P-201: start against HV-263 throttled to ≤ 60 m³/h (FI-284) until V-301 N-F is covered, then open fully; stop on LIT-2k7 low of the tank being emptied.

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 07 — PS-02 Heater E-201, P-202, P-201

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

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Ref.: PS02-DB-001 Rev P1 (D-E … D-P, D-X)

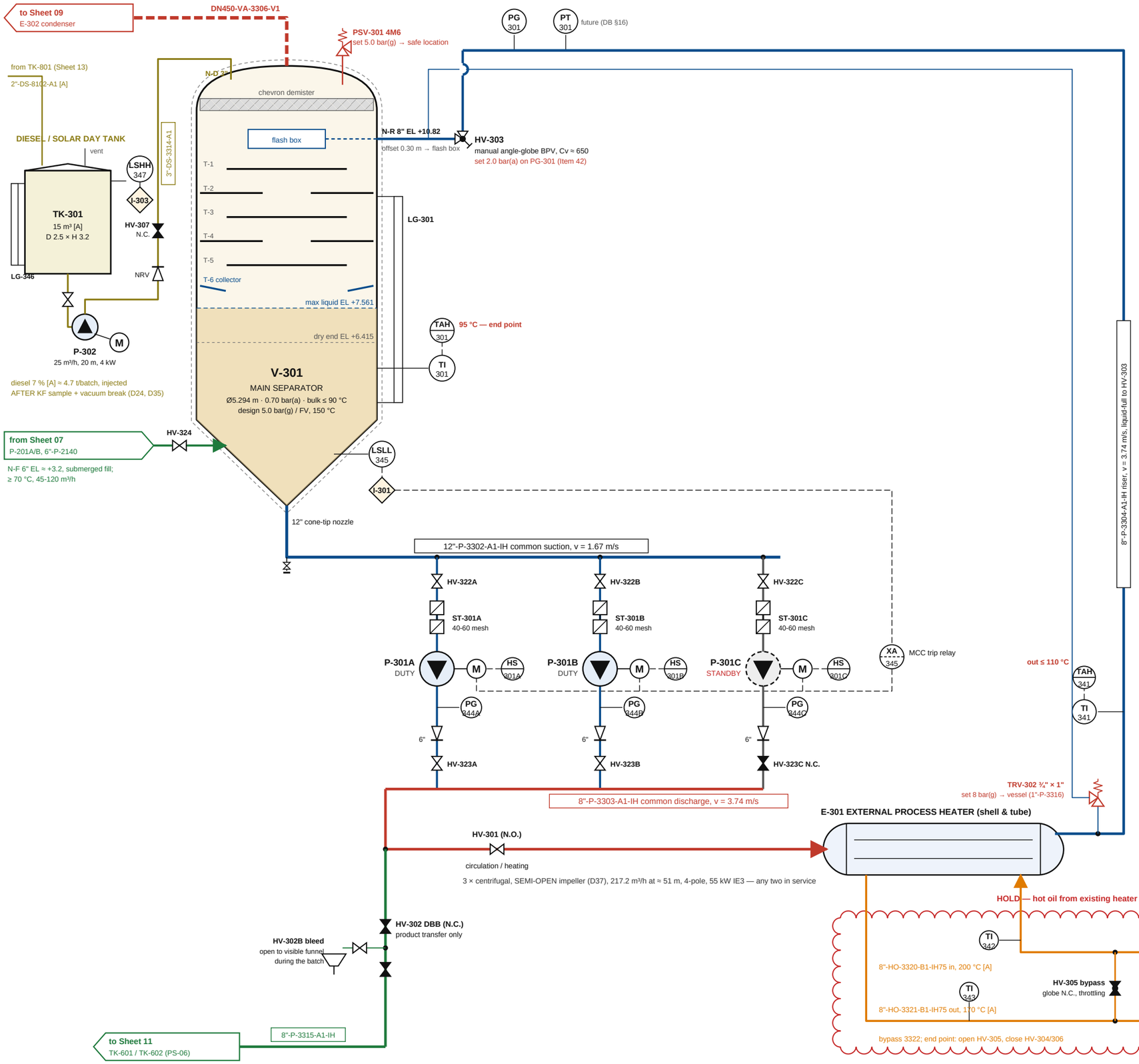
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Sheet 07 of 15

A3 — NTS (schematic)

PS-03 — MAIN SEPARATOR V-301, HEATER LOOP E-301, PUMPS P-301A/B/C, DIESEL INJECTION, PRODUCT TRANSFER

P&ID, NOT TO SCALE. Basis: PS03-DB-001 Rev P18 (Sections 7, 11, 15-17; D24-D38) and PS03-PID-001 Rev P9 Sheet 1 (superseded by this sheet). EL in m above grade. Vacuum & condensing: Sheet 09.



TRIP / INTERLOCK AND ALARM SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-301	LSLL-345 cone	0.5 m above outlet [A]	Stop P-301A/B/C	Manual LCS
TAH-301	TI-301 bulk	95 °C	Alarm: end point	Self
TAH-341	TI-341 E-301 out	110 °C	Alarm	Self
I-303	LSHH-347 TK-301	90 % [A]	Stop P-802 (Sh.13)	Manual

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-301A/B/C	55 IE3	soft-start [A]	LCS HS-301A/B/C	I-301
P-302	4	DOL	LCS	—

KEY DATA — SHEET 08

Item	Value
Charge / water removed	89.7 t wet / 22.2 t per batch
Design / normal duty	4.14 / 3.52 MW
Hot oil supply / return	200 / 170 °C [A]
Circulation (2 pumps)	434.3 m³/h, ΔT ≈ 20 °C per pass
Back-pressure at HV-303	2.0 bar(a) ≈ 1.0 bar(g)
NPSH available	3.98 m (cone-tip suction)
Flash box / holes	Ø1.40 × 0.50 m / 100 × Ø50 mm
Internals	T-1…T-5 + T-6 collector (LAY P4)
E-301	2" Sch40, 22 × 6 passes, 241.5 m², 2.5 m/s
E-301 U / ΔP / design	163-226 W/m²K / 2.17 bar / 8 bar(g), 230 °C
Insulation	V-301, E-301, loop: 50 mm (D33)

NOTES

1. PS-03 has only three pumps (any two in service, single phase).
2. HV-303 throttling = the calculated shear (d_crit 18-38 µm); the flash happens in the flash box, not in E-301 or the riser.
3. End of dehydration (D28): brine-plateau exit or TAH-301 95 °C → open HV-305, close HV-304/306; P-301 keep circulating; KF / D95 sample BEFORE diesel (D24). Heater isolated on hot-oil side only.
4. Loss of a duty pump: start P-301C and open HV-305 at once.
5. Product transfer (vessel at atmosphere): HV-301 closed, HV-302 DBB open, HV-302B closed. During the batch HV-302 closed with HV-302B open to the funnel (any passing is seen at once).
6. Diesel AFTER the vacuum break: P-302 → HV-307 → N-D (D35).
7. Renumbered from P9: E-1 → E-301, P-3A/B/C → P-301A/B/C, P-4 → P-302, TK-2 → TK-301, HV-1…7 → HV-301…307, stream 3xx → line 33xx.
8. New on this sheet: LSLL-345/I-301, TAH-301, TAH-341, HV-322/323, HV-324 at N-F, PG-344A/B/C, TI-342/343, LG-346, LSHH-347/I-303.

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 08 — PS-03 V-301, E-301, P-301A/B/C

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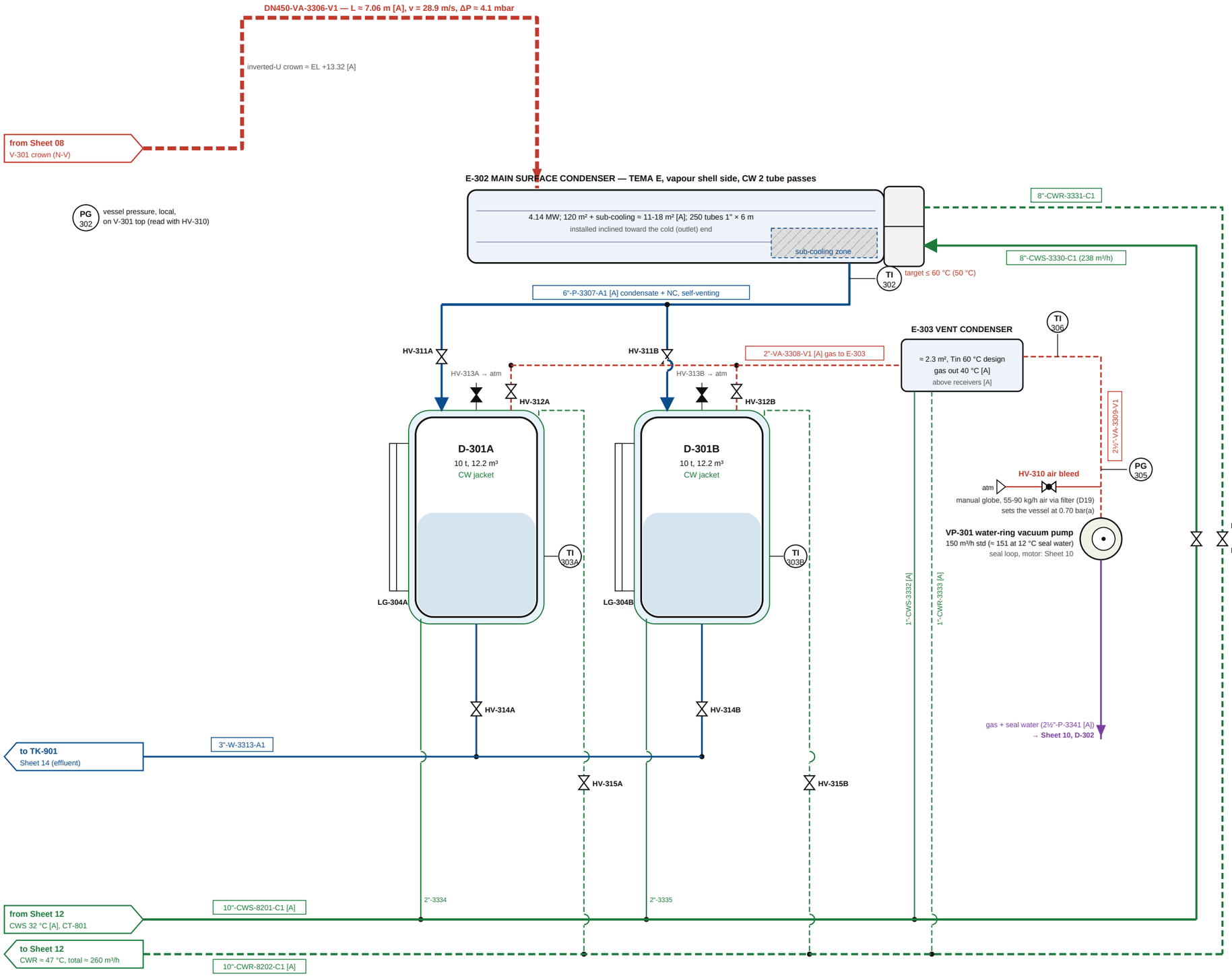
Sheet 08 of 15

A3 — NTS (schematic)

Supersedes PS03-PID-001 P9 Sh.1

PS-03 — VACUUM AND CONDENSING SYSTEM: E-302, RECEIVERS D-301A/B, VENT CONDENSER E-303, VACUUM PUMP VP-301

P&ID, NOT TO SCALE. Basis: PS03-DB-001 Rev P18 Section 10 and PS03-PID-001 Rev P9 Sheet 2 (superseded by this sheet). VP-301 seal-water loop: Sheet 10. Cooling water from CT-801: Sheet 12.



KEY DATA — SHEET 09

Item	Value
Design vapour load	1.813 kg/s (6.53 t/h), 4.35 m³/s
Vessel / condensing / suction	700 / 690 / 680 mbar(a)
E-302	TEMA E, 4.14 MW, 120 m² design
E-302 sub-cooling [A]	+225 kW, 11-18 m², out ≤ 60 °C
E-302 cooling water	238 m³/h, 32 → 47 °C
D-301A/B	2 × 10 t (12.2 m³), alternating
Receiver jackets	≈ 10 m³/h each [A]; water ≈ 50-53 °C
Gravity drain after vacuum break	≈ 12 min (3"), 6.6 min (4")
E-303	≈ 2.3 m², Tin 60 °C, out 40 °C
VP-301	150 m³/h (≈ 151 at 12 °C seal)
Process suction flow	59 m³/h (NC 40 kg/h [A])
Air bleed HV-310	55-90 kg/h on pump suction
Cooling water total	≈ 260 m³/h (with jackets)
Design	E-302 shell, D-301, E-303, VA lines: 5.0 bar(g) / FV

TRIP / INTERLOCK AND MOTOR DATA

Trips	none (manual vacuum control by HV-310)
Motors	VP-301: see Sheet 10

NOTES

1. Protected against over-pressure by PSV-301 on V-301 (Sheet 08) (loss of cooling governs, D34).
2. Receivers not thermally insulated (personnel protection only).
3. E-303 is mounted above the receivers; its condensate returns by gravity through the gas line [A].
4. Condensate quality / destination: Item 20 (TQ-004).
5. Renumbered from P9: E-2 → E-302, D-1A/B → D-301A/B, E-3 → E-303, VP-1 → VP-301, HV-10…15 → HV-310…315; streams 306-335 → lines 3306-3335. New: HV-325/326 (E-302 CW).
6. CW headers sized 10" [A] for ≈ 260 m³/h (v ≈ 1.4 m/s).

PRESSURE BUDGET AND OPERATION (DB 10.1, D19, D20, D28)

Vessel 700 mbar(a) → condensing zone 690 mbar(a) → VP-301 suction 680 mbar(a); Tsat 89.9 → 89.6 → 89.2 °C. Vessel pressure is set by HV-310 (manual bleed on the pump suction), read on PG-302; the bleed never passes through E-302 / E-303 and keeps the headspace air-lean. Receiver water temperature governs the vapour carried to E-303 (89 °C ≈ 1,166 kg/h; 60 °C 10 kg/h; 50 °C 5 kg/h).
Receiver switching (≈ every 1.5 h): close HV-311/HV-312 of the full receiver, open HV-313 (atmosphere), drain by HV-314; close HV-314/HV-313, re-open HV-312 slowly (≈ 14 kg air, trim HV-310), then HV-311.
Condensate and non-condensables leave E-302 together through the bottom (coldest) nozzle and drain by gravity to the receivers; the receiver top is the gas off-take (E-303 first, then VP-301).

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 09 — PS-03 Vacuum and condensing

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Rev: P1 Status: IFR

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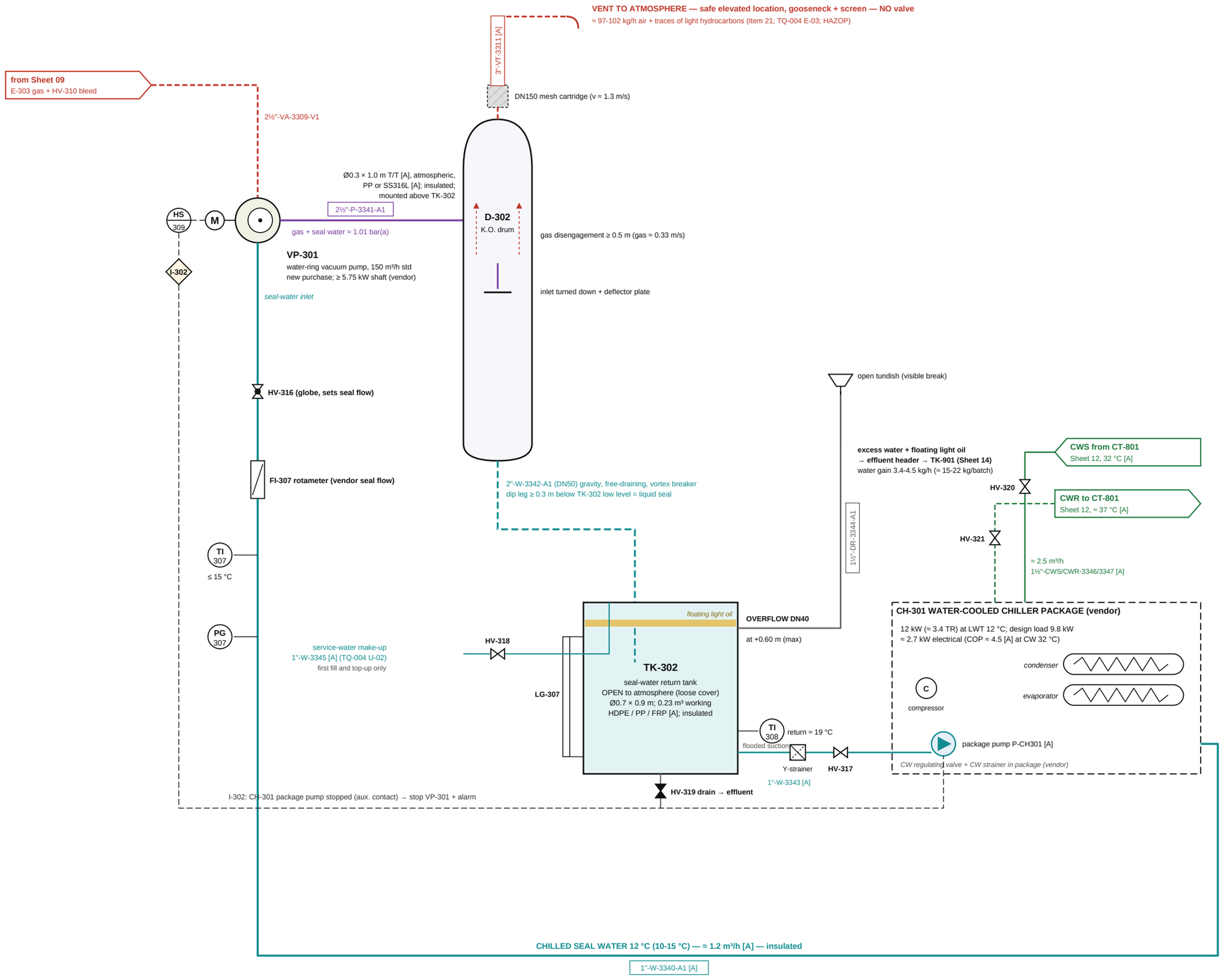
Sheet 09 of 15

A3 — NTS (schematic)

Supersedes PS03-PID-001 P9 Sh.2

PS-03 — VP-301 SEAL-WATER CHILLED LOOP: KNOCK-OUT DRUM D-302, RETURN TANK TK-302, WATER-COOLED CHILLER CH-301

P&ID, NOT TO SCALE. Basis: PS03-DB-001 Rev P18 Section 10.11, KH CAL-001 K.9 and PS03-PID-001 Rev P9 Sheet 3 (superseded by this sheet). Vacuum system: Sheet 09. Cooling water: Sheet 12.



TRIP / INTERLOCK SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-302	CH-301 pump run	pump stopped	Stop VP-301 + alarm	Manual LCS

MOTOR DATA

Tag	kW	Starter	Control	Trip by
VP-301	7.5 [A]	DOL	LCS HS-309	I-302
CH-301 package	≈ 2.7 + pump	package	package panel	—

KEY DATA — SHEET 10

Item	Value
Cooling duty, design / normal	9.8 / ≈ 7 kW
CH-301 (water-cooled)	12 kW (≈ 3.4 TR), LWT 12 °C
Heat rejected to CT-801	≈ 14.7 kW at 12 kW duty
CW branch	≈ 2.5 m³/h, 32 → 37 °C [A]
Seal water to VP-301	≈ 1.2 m³/h [A], 12 → ≈ 19 °C
Water gain of the loop	3.4-4.5 kg/h → overflow
D-302	Ø0.3 × 1.0 m T/T [A], atmospheric
TK-302	Ø0.7 × 0.9 m, 0.23 m³ working
Vent gas	≈ 97-102 kg/h air + traces
Insulation	all cold lines, D-302, TK-302 (dew point)

NOTES

1. Closed loop: CH-301 → VP-301 seal inlet → VP-301 discharge → D-302: gas to atmosphere; water by gravity to TK-302 → CH-301.
2. No valve on the D-302 gas outlet / vent line (a blocked vent would pressurise D-302 / TK-302).
3. TK-302 overflow at its highest liquid point removes the excess water with the floating light oil, via an open tundish.
4. Seal flow set with HV-316 on FI-307; TI-307 ≤ 15 °C.
5. I-302 (new): a liquid-ring pump without seal water is damaged in minutes — VP-301 stops when the package pump stops.
6. Renumbered from P9: VP-1 → VP-301, D-2 → D-302, TK-1 → TK-302, CH-1 → CH-301, HV-16…21 → HV-316…321; lines 3311, 3340-3347.

COOLING-LOAD BALANCE AND SIZING (DB 10.11; KH K.9)

Q = P_shaft(VP-301) + Q_vapour condensed + Q_NC + Q_bleed air + Q_gains = 5.75 + 3.0 + 0.3 + 0.3 + 0.5 ≈ 9.8 kW (normal ≈ 7 kW). CH-301 condenser: Q_rej = 12 + 12/4.5 ≈ 14.7 kW → CW ≈ 2.5 m³/h, 32 → 37 °C [A].

D-302 (Souders-Brown): v_max = 0.05·√((998 − 1.2)/1.2) ≈ 1.4 m/s → Ø0.3 m chosen for inlet-jet dissipation; gas ≈ 0.33 m/s → gravity cut ≈ 0.12 mm; DN150 mesh cartridge 1.3 m/s.

TK-302: downflow Q/A = 0.87 mm/s = Stokes rise of an ≈ 85 µm light-oil drop → larger drops float and leave by the overflow; buffer 230 L (≈ 19 L per kW of chiller capacity, ≈ 11.5 min of seal flow).

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 10 — PS-03 VP-301 seal-water loop, CH-301

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

Author: Eng. Mohamed Tarek
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Ref.: PS03-DB-001 Rev P18 §10.11; KH CAL-001 K.9

Date: 19-Oct-2026

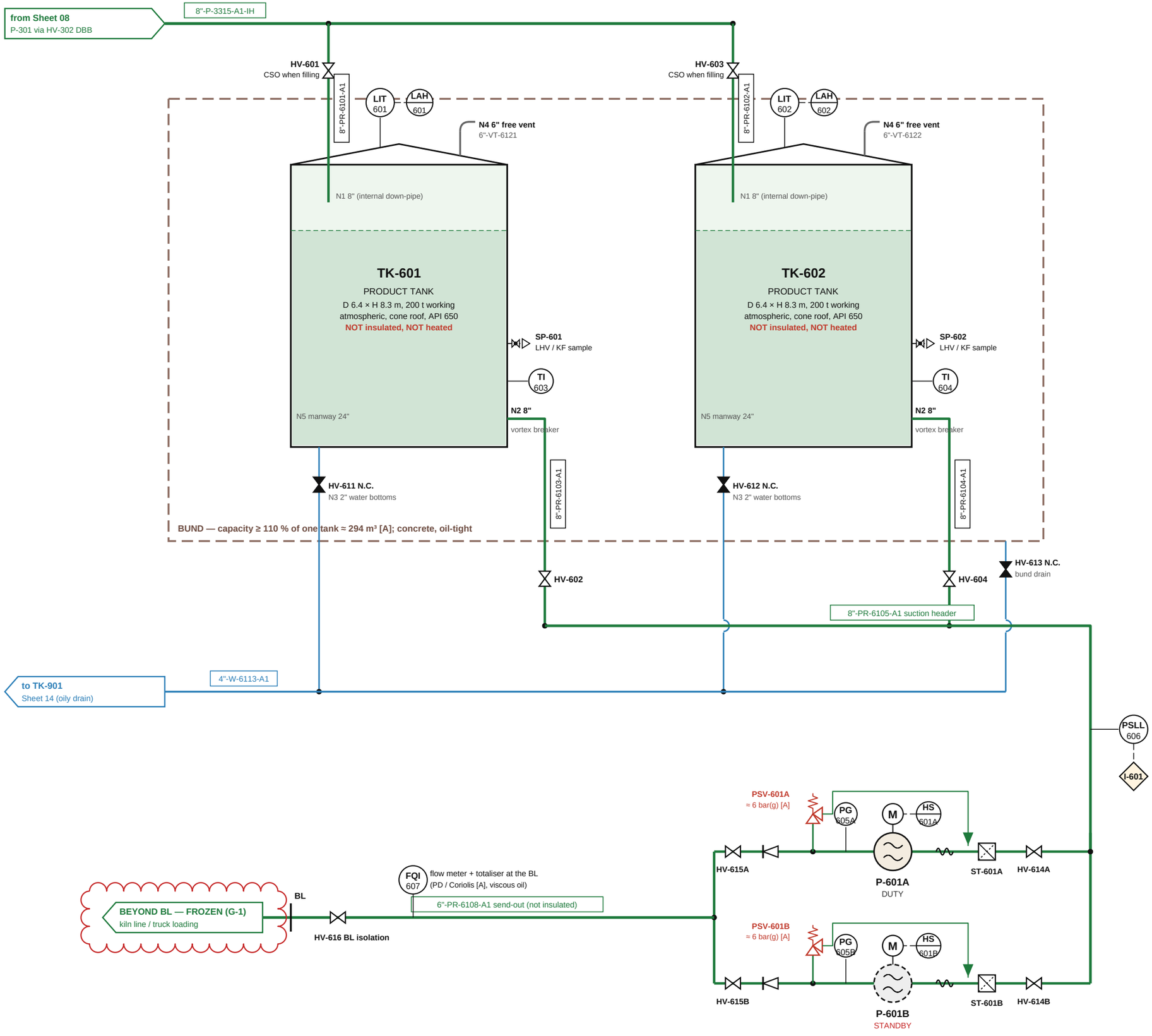
Sheet 10 of 15

A3 — NTS (schematic)

Supersedes PS03-PID-001 P9 Sh.3

PS-06 — PRODUCT STORAGE TK-601 / TK-602 AND SEND-OUT PUMPS P-601A/B

P&ID, NOT TO SCALE. Basis: PS-06 closure 16-Oct-2026 (2 × 200 t, H/D ≈ 1.3), GEN-EL-001 Rev P2, G-1 (send-out beyond the battery limit FROZEN). Tanks NOT insulated and NOT heated (Author 18-Oct-2026).



TRIP / INTERLOCK AND ALARM SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-601	PSLL-606	0.1 bar(g) [A]	Stop P-601A/B	Manual LCS
LAH-601/602	LIT-601/602	90 % [A]	Alarm (stop filling)	Self

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-601A/B	11 [A] HOLD	DOL *	LCS HS-601A/B	I-601

** PD pump: direct-on-line, not star-delta.*

KEY DATA — SHEET 11

Item	Value
Product rate	216.5 t/day (66.7 oil + 4.7 diesel per batch × 3)
Storage	2 × 200 t ≈ 1.8 days
Fill	per PS-03 batch, ≈ 72 t, by P-301 (≈ 217 m³/h)
Temperature	≈ 90 °C in, cools to ambient (no insulation)
Cold viscosity	dry oil ≈ 1.6 Pa·s at 10 °C (less with diesel)
Send-out line	6": 0.5-0.8 bar per 50 m cold (30-50 m³/h)
Vent N4 6"	out ≈ 220 Nm³/h at fill → ΔP < 0.05 kPa
Design	atmospheric, ambient … 95 °C, class A1

NOTES

1. Tanks and lines NOT insulated, NOT heated (Author 18-Oct).
Personnel protection (PP) only where the hot fill line is within reach (≈ 90 °C during a transfer).
2. One tank receives, the other sends out; HV-601/603 inlet valves car-sealed per procedure (as PS-02).
3. Water bottoms: drain N3 before each send-out to TK-901.
4. Bund drain HV-613 N.C.: rain water checked for oil before draining; oily water to TK-901.
5. Beyond the battery limit (destination, loading method, rate) FROZEN per G-1; P-601 duty HOLD until fixed.
6. Product quality (LHV, KF water) sampled at SP-601/602.

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 11 — PS-06 Product storage and send-out

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

Author: Eng. Mohamed Tarek
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Ref.: PS-06 closure 16-Oct; GEN-EL-001 Rev P2; G-1

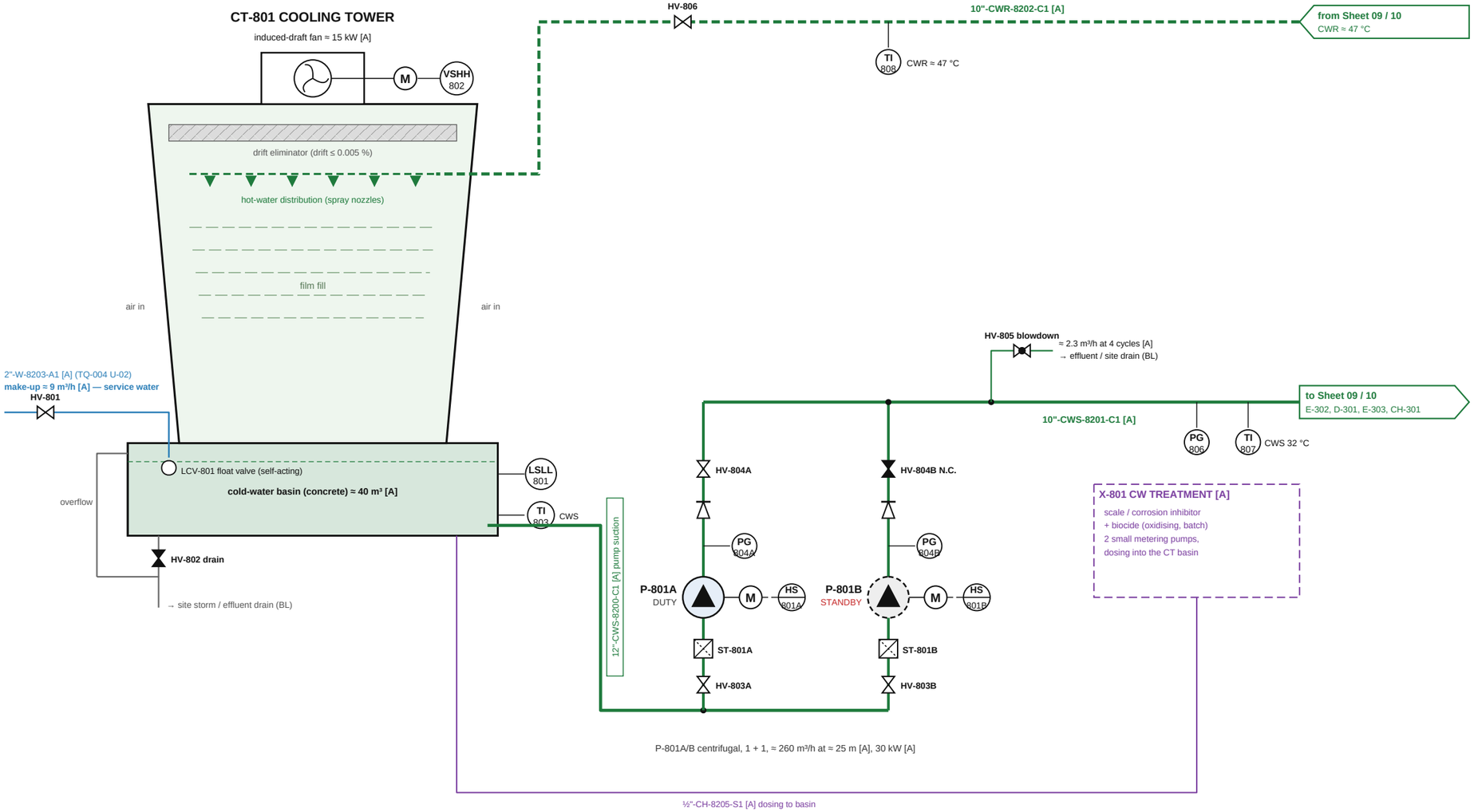
Date: 19-Oct-2026

Sheet 11 of 15

A3 — NTS (schematic)

UTILITIES — COOLING TOWER CT-801, CW PUMPS P-801A/B AND COOLING-WATER NETWORK (CONDENSERS ONLY)

P&ID, NOT TO SCALE. Basis: G-5 (own cooling tower in our scope, condensers only), PS03-DB-001 Rev P18 §10 (CW 238 m³/h, 32 → 47 °C), Sheet 09 (E-302, D-301A/B jackets, E-303), Sheet 10 (CH-301).



TRIP / INTERLOCK SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-801	LSLL-801 basin	low-low [A]	Stop P-801A/B	Manual LCS
I-802	VSHH-802 fan	high-high vib.	Stop CT fan	Manual LCS

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-801A/B	30 [A]	soft-start [A]	LCS HS-801A/B	I-801
CT-801 fan	15 [A]	DOL / VFD [A]	LCS	I-802
X-801 pumps	2 × 0.18 [A]	package	package	—

KEY DATA — SHEET 12

Item	Value
Heat load (design)	≈ 4.5 MW [A]
Circulation	≈ 260 m³/h
CWS / CWR	32 / ≈ 47 °C (range ≈ 15 K)
Design wet bulb	≈ 26-27 °C summer [A] → approach 5-6 K
Evaporation	≈ 6.8 m³/h (Q / h_fg)
Blowdown	≈ 2.3 m³/h at 4 cycles [A]
Make-up	≈ 9 m³/h service water
Type	induced draft, counterflow, 1 cell [A]

NOTES

1. Own cooling tower, condensers only (G-5); heating source stays the existing hot-oil system.
2. Water balance: make-up = evaporation + blowdown + drift; E = Q / h_fg ≈ 4.5 MW / 2,400 kJ/kg ≈ 1.9 kg/s ≈ 6.8 m³/h; B = E / (C − 1) at C = 4 [A] ≈ 2.3 m³/h.
3. Basin level held by the self-acting float valve LCV-801; I-801 protects P-801 against running dry.
4. Blowdown set by hand on HV-805 from weekly conductivity checks [A]; treatment chemicals per vendor (X-801).
5. P-801 rating 30 kW [A] supersedes the earlier ≈ 22 kW estimate (E-302 is mounted high; TDH ≈ 25 m [A]).

CW CONSUMERS (branches on Sheets 09 and 10)

Consumer	Sheet	Flow	In / out	Duty	Branch lines
E-302 main condenser	09	238 m³/h	32 → 47 °C	4.14 MW + 0.23 sub-cooling	8"-CWS-3330 / 8"-CWR-3331
D-301A/B receiver jackets (alternating)	09	≈ 2 × 10 m³/h [A]	32 → ≈ 37 °C	condensate 50-53 °C	2"-3334 / 2"-3335
E-303 vent condenser	09	≈ 1 m³/h [A]	32 → 40 °C	≈ 30 kW	1"-CWS-3332 / 1"-CWR-3333
CH-301 chiller condenser	10	≈ 2.5 m³/h	32 → 37 °C	14.7 kW	1½"-3346 / 1½"-3347
TOTAL (design)		**≈ 260 m³/h**		**≈ 4.5 MW [A]**	headers 10" [A], v ≈ 1.4 m/s

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 12 — Cooling tower CT-801 and CW network

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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Ref.: G-5; PS03-DB-001 Rev P18 §10; Sheets 09, 10

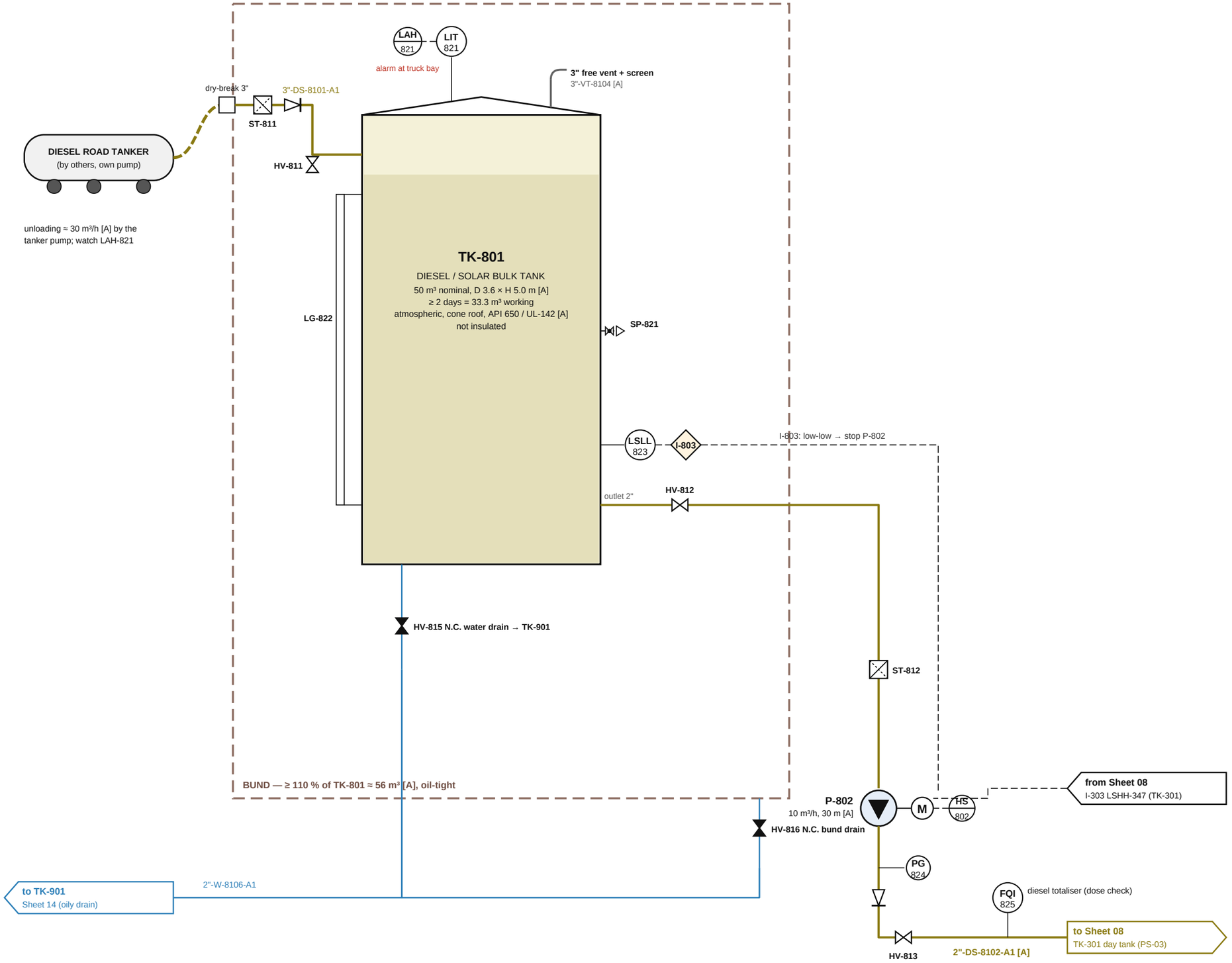
Date: 19-Oct-2026

Sheet 12 of 15

A3 — NTS (schematic)

UTILITIES / OFFSITES — DIESEL (SOLAR) BULK STORAGE TK-801, TRUCK UNLOADING AND TRANSFER PUMP P-802 TO PS-03

P&ID, NOT TO SCALE. Basis: G-4 (≈ 50 m³ plain tank, ≥ 2 days, bunded, small transfer pump), G-8 (TK-801 in the utilities area, own bund), PS03-DB-001 Rev P18 D38. Diesel 7 % [A] ≈ 4.7 t per PS-03 batch ≈ 14 t/day.



TRIP / INTERLOCK AND ALARM SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-803	LSLL-823 TK-801	low-low [A]	Stop P-802	Manual LCS
I-303	LSHH-347 TK-301	90 % [A]	Stop P-802	Manual
LAH-821	LIT-821	90 % [A]	Alarm at truck bay	Self

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-802	2.2 [A]	DOL	LCS HS-802	I-803, I-303

KEY DATA — SHEET 13

Item	Value
Use	≈ 4.7 t per PS-03 batch, ≈ 14 t/day
TK-801	50 m³ nominal, ≥ 2 days
TK-301 day tank	15 m³ [A] (Sheet 08)
P-802	centrifugal, 10 m³/h at 30 m [A]
Fill of TK-301	≈ 1.5 h for 15 m³
Flash point	Egyptian solar ≥ 55 °C [A], handled ambient
Class	A1; not insulated

NOTES

1. TK-801 in the utilities / offsites area with its own bund (G-8).
2. Truck unloading by the tanker's own pump through a 3" dry-break coupling, strainer and NRV; operator watches LIT-821 / LAH-821.
3. P-802 tops up TK-301 before each injection; I-303 (LSHH-347 on TK-301, Sheet 08) prevents an overfill — diesel spill risk.
4. Water bottoms drained to TK-901 via HV-815 (weekly [A]).
5. Motors ordinary (unclassified area, Author 18-Oct); keep ignition sources out of the bund per Civil Defence rules.

SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 13 — Diesel storage TK-801 and transfer

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-PID-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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Ref.: G-4, G-8; PS03-DB-001 Rev P18 D38

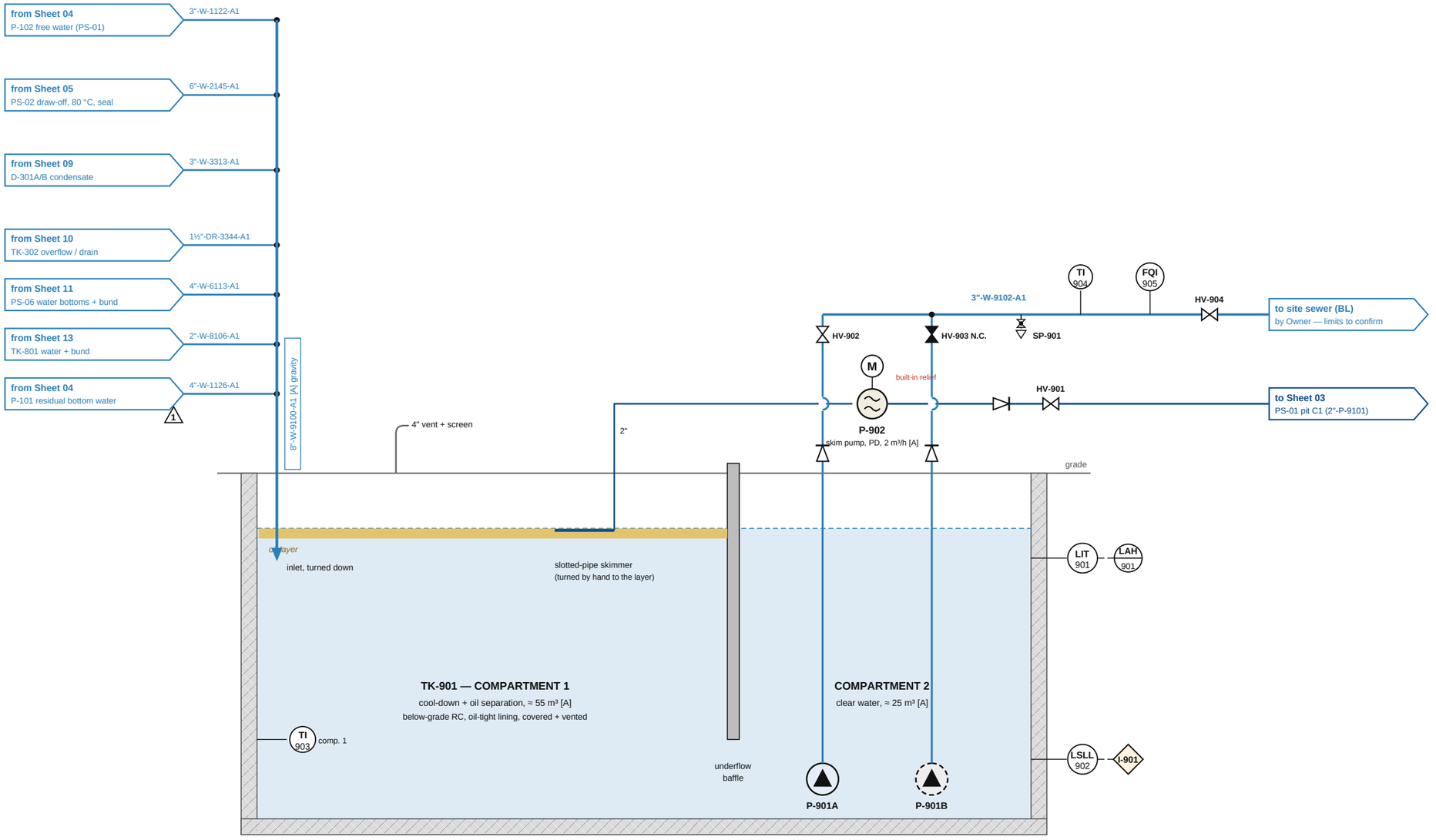
Date: 19-Oct-2026

Sheet 13 of 15

A3 — NTS (schematic)

EFFLUENT — COOL-DOWN / OIL-SKIM TANK TK-901, DISCHARGE PUMPS P-901A/B, SKIM PUMP P-902, SLUDGE SKIP SK-901

P&ID, NOT TO SCALE. Basis: G-2 (all water streams to TK-901: gravity cool-down + oil skim; skimmed oil back to PS-01 pit; sludge disposal later), whole-train balance 17-Oct (≈ 205 m³/day water). Discharge limits to be confirmed.



P-901A/B submersible, 30 m³/h at 15 m [A], 3 kW

TRIP / INTERLOCK AND ALARM SCHEDULE

Tag	Initiator	Setpoint	Action	Reset
I-901	LSLL-902 comp. 2	low-low [A]	Stop P-901A/B	Manual LCS
LAH-901	LIT-901	high [A]	Alarm	Self

MOTOR DATA

Tag	kW	Starter	Control	Trip by
P-901A/B	3 [A]	DOL	LCS HS-901A/B	I-901
P-902	0.75 [A]	DOL	LCS HS-902	—

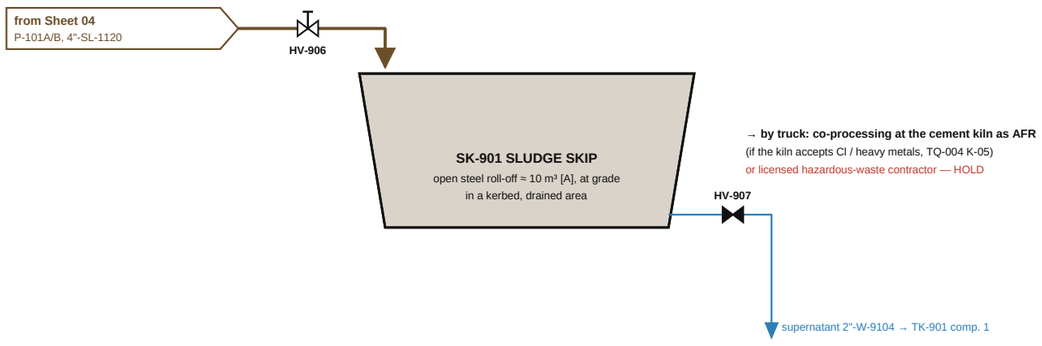
KEY DATA — SHEET 14

Item	Value
Water to TK-901	≈ 205 m³/day (≈ 8.5 m³/h mean)
Largest slug	PS-02 draw-off 18.6 m³ at 80 °C in ≈ 8 min
Cool-down heel	≥ 53 m³ at 30 °C → mix ≤ 43 °C
TK-901 working	≈ 80 m³ [A] (55 + 25), below grade
Salinity	≈ 37 g/kg (sea-water level)
Discharge	batch, ≈ 30 m³/h, after T and sample check
Limits (to confirm)	O&G ≈ 100 mg/L, T ≤ 43 °C, pH 6-9.5

NOTES

1. Discharge only when TI-904 ≤ 43 °C and SP-901 shows no free oil (operating manual). Limits per Law 93/1962, Decree 44/2000 — to be confirmed with the receiving authority.
2. Before a PS-02 draw-off, compartment 1 must hold its cool-water heel (≥ 53 m³): do not pump TK-901 down below it.
3. Skimmed oil returns to the PS-01 pit and is re-processed.
4. High salinity (≈ sea water) — confirm the sewer accepts it (TQ-004 L-18 conductivity).
5. Sludge disposal route HOLD (G-2): kiln AFR or licensed contractor; sludge quantity by TQ-004 L-17 / D473 / D4007.
6. CT-801 blowdown / overflow go to the site drain, not TK-901.

SLUDGE (PS-01 P-101A/B)



SEVERO WM OIL PROJECT

Plant P&ID Set — Process & Instrumentation Diagrams

Sheet 14 — Effluent TK-901 and sludge skip

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Class CL — Client Deliverable

Rev: P1 Status: IFR

Author: Eng. Mohamed Tarek
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Ref.: G-2; PS01-DB-001 Rev.2 §10; whole-train balance 17-Oct

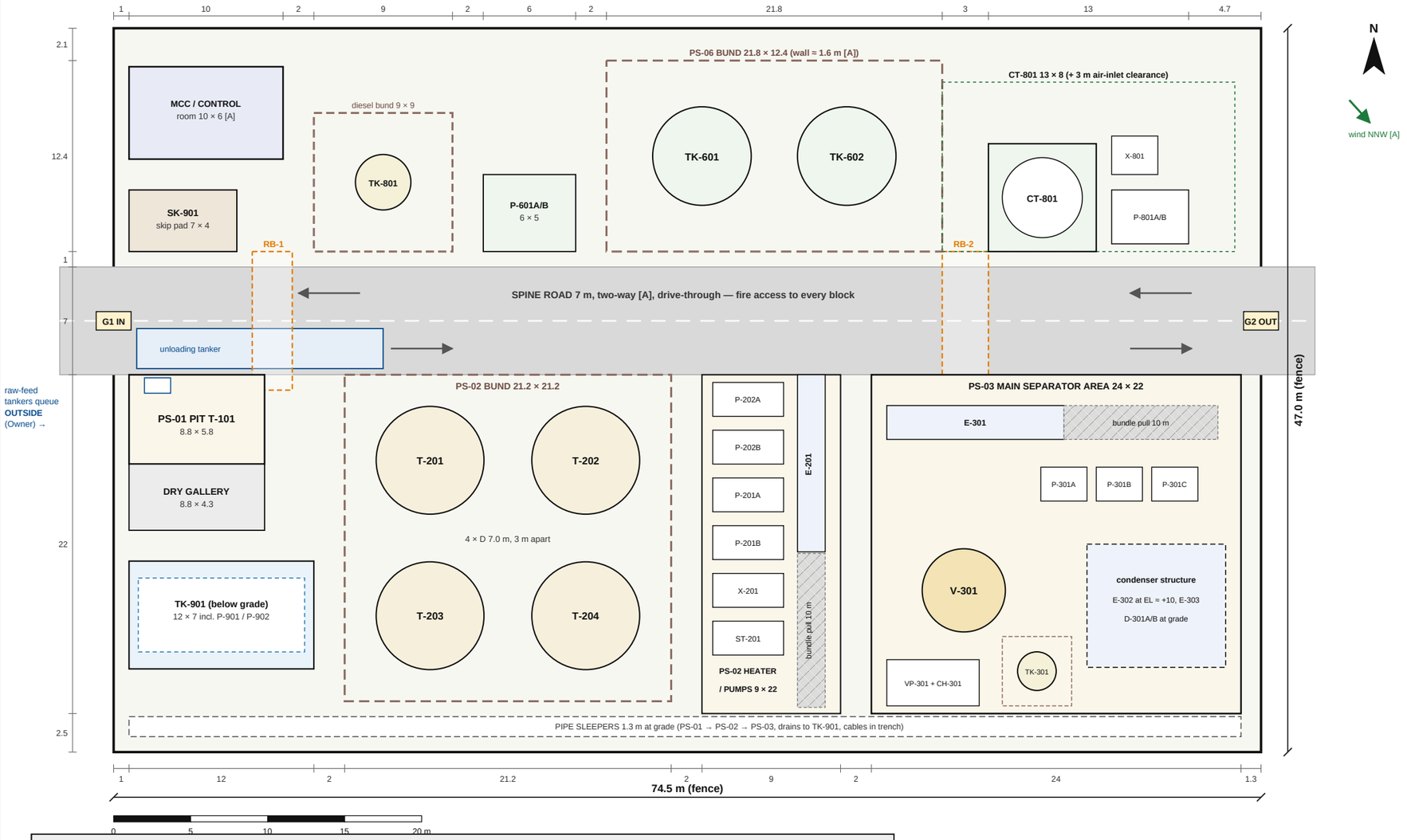
Date: 19-Oct-2026

Sheet 14 of 15

A3 — NTS (schematic)

SITE GENERAL ARRANGEMENT / PLOT PLAN — PRELIMINARY (NO SITE BOUNDARY) AND REQUIRED PLOT AREA

Rev P1: compact two-row layout on one spine road (owner 19-Oct: minimum plot). Basis: GEN-PID-001 Rev P1, unit DBs, G-5, G-7, G-8, NFPA 30 Table 22.4.1.6. Dimensions in m. Scale ≈ 1:320 on A3.



REQUIRED PLOT AREA

Item	Value
Fenced rectangle	74.5 × 47.0 m
Area	3,502 m² ≈ 0.35 ha
Recommended plot	≥ 76 × 49 m ≈ 0.37 ha
Equipment blocks	2,016 m² (58 %)
Spine road	≈ 522 m² (15 %)
Rev P0 (ring road)	123 × 73 m = 8,979 m²

KEY

Symbol	Meaning
grey band	spine road 7 m
brown dashed	bund wall
blue hatched	bundle-pull space
dashed box	structure / clearance
orange dashed	rack bridge RB over the road
[A]	assumed, to confirm

NOTES

1. Preliminary layout for area estimate and Owner discussion; final positions after the site, soil and Civil Defence data.
2. PS-01 pit at the road edge: tanker unloads from the road into BSK-101A/B (grade-flush baskets).
3. TK-901 behind PS-01 and next to PS-02: PS-02 tundish drain by gravity; PS-02 overflow N10 → PS-01 pit by gravity.
4. CT-801 opposite PS-03 (10" CW over RB-2, ≈ 20 m).
5. SK-901 opposite PS-01 (P-101 sludge 4" over RB-1).
6. MCC at the NW corner: up-wind and ≥ 38 m from PS-03 (only unit with light ends / vacuum vent).
7. P-601A/B next to PS-06 at the road (product bay G-1 FROZEN).

PLOT AREA CALCULATION (m²) — Rev P1 compact layout

Block	Size (m)	Area (m²)
PS-01 pit + dry gallery	8.8 × 10.1	89
SK-901 sludge skip pad	7.0 × 4.0	28
TK-901 effluent (below grade) + pumps	12.0 × 7.0	84
PS-02 tank bund (4 × D 7.0)	21.2 × 21.2	449
PS-02 heater / pumps (E-201 + 10 m pull inside)	9.0 × 22.0	198
PS-03 main separator area	24.0 × 22.0	528
PS-06 product bund (2 × D 6.4)	21.8 × 12.4	270
P-601A/B pad	6.0 × 5.0	30
CT-801 area (excl. air clearance)	13.0 × 8.0	104
TK-801 diesel bund	9.0 × 9.0	81
MCC / control room	10.0 × 6.0	60
Pipe sleepers (rack bridges over the road)	72.2 × 1.3	94
Sum of equipment blocks		**2,016**
Spine road 7 m	74.5 × 7.0	522
Spacing, verges, maintenance access		964
FENCED PLOT (rectangle)	**74.5 × 47.0 m**	**3,502 m² ≈ 0.35 ha**
Rev P0 (ring road, 123 × 73 m)		8,979 → −61 %

BASIS AND ASSUMPTIONS (Rev P1, owner 19-Oct: minimum plot area)

1. Two rows on ONE spine road 7 m (drive-through G1 → G2): process row south (PS-01 → PS-02 → PS-03, gravity links kept), storage / utility row north. Every block fronts the road.
2. Raw-feed tanker queue OUTSIDE the fence (Owner). One tanker unloads at PS-01 on the road.
3. Spacing ≥ 2 m between blocks (3 m CT air inlet), 1 m verges (2.1 m north: PS-06, note 4).
4. NFPA 30 Table 22.4.1.6 (Class IIIB) shell → property line: PS-02 (33,000 gal) ≥ 3.05 m → 5.4 m; PS-06 (70,500 gal) ≥ 4.6 m → 5.1 m; TK-801 ≥ 3.05 m. Shell → MCC ≥ 3.05 m. Met.
5. Bunds ≥ 110 % of the largest tank: PS-02 137.5 m³, PS-06 294 m³, TK-801 56 m³.
6. E-201 / E-301 bundle pulls stay INSIDE their blocks (road never blocked).
7. Rack bridges RB-1 / RB-2: clear height ≥ 6.0 m over the road [A] (tanker ≈ 4.3 m).
8. If the site allows access from the west only: east turning head ≈ 18 × 14 m (+≈ 250 m²).
9. Not included: hot-oil heater (tie-in by Owner), fire water, admin / lab, parking.
10. **Final spacing to Egyptian Civil Defence rules (Owner). RECOMMENDED PLOT ≥ 76 × 49 m.**

SEVERO WM OIL PROJECT

Site General Arrangement / Plot Plan

Preliminary — no site boundary; required plot area

Doc. No.: SEVERO-WMO-CL-ENG01-GEN-LAY-001

Class CL — Client Deliverable

Rev: P1 Status: IFR

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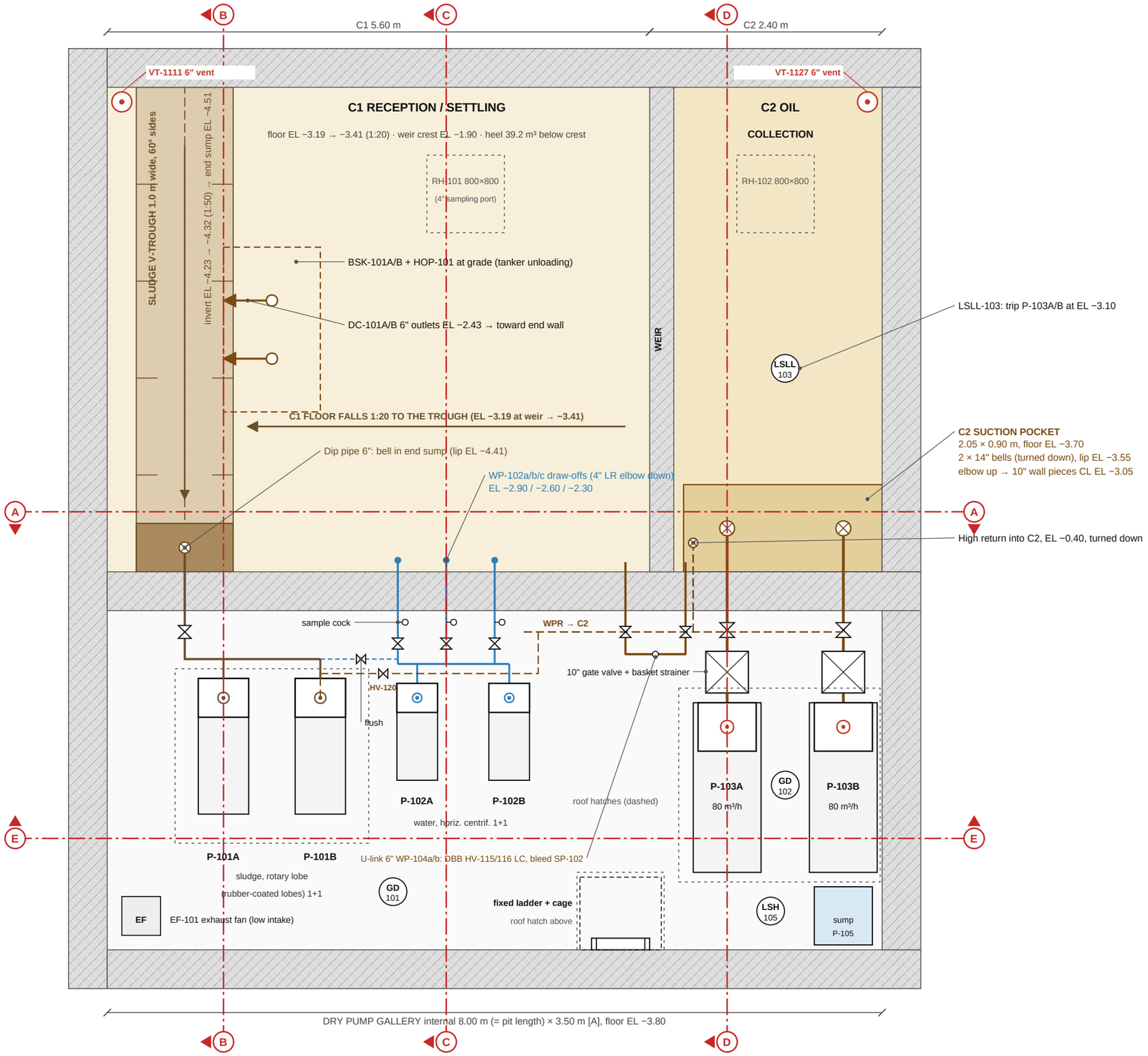
Sheet 1 of 1

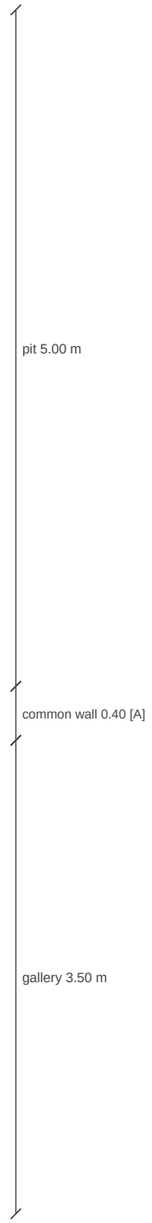
A3 — NTS (schematic)

Issued in the same PDF as GEN-PID-001

PS-01 — RECEPTION PIT + DRY PUMP GALLERY — SHEET 1: PLAN

CONCEPT / IFR — drawn to scale ratio (NTS). Basis: SEVERO-WMO-CL-ENG01-PS01-DB-001 Rev.2 (Rev.2: floor falls, weir 1.40 m, U-link + DBB, P-101 → C2, hatches 800 × 800, two vents, gallery drainage).





KEY DATA

Pit internal / mean floor	**8.0 × 5.0 m / EL −3.30**
Batch / max level	**100 t / EL −0.52 (C1 emptied)**
Weir / crest (Rev.2)	**1.40 m / EL −1.90**
Gallery internal [A]	**8.0 × 3.5 m (= pit length)**
Gallery floor	**EL −3.80**
P-103A/B oil	**80 m³/h lobe, DOL, 10" suct.**
P-101A/B sludge	**lobe — HOLD duty; → skip / TK-901 / C2**
P-102A/B water	**centrif. ≥ 40 m³/h, runs while unloading**
Fill line to PS-02	**8" (discharge risers)**
C2 pump trip (LSLL-103)	**EL −3.10**
Roof hatches / vents	**800 × 800 / 2 × 6", 1.5 m**

NOTES

1. Dimensions in m; EL from grade 0.00; NTS.
2. All gallery footprints [A] until vendor data.
3. Weir keeps sludge in C1; sludge pulled from the V-trough end sump each batch (order: sludge → water → oil).
4. Water drawn from the lowest draw-off showing water; sample cock on each wall piece shows oil/emulsion/water.
5. P-102 runs from WP-102a during unloading (M-12); discharge to TK-901, or rag layer to C2 via WPR. WPR (high level) also takes PSV-103 and the C1 residual.
6. All wall pieces cast-in with puddle flange.
7. Gallery = confined space: EF-101, GD-101/102, LSH-105.
8. C1 floor 1:20 to the trough; wash-down hose points; vacuum truck backup through RH-101 / RH-102.
9. Access: fixed vertical ladder with safety cage (fall height ≈3.8 m > 3 m, ISO 14122-4) + roof hatch.
10. Orientation / north: see site GA GEN-LAY-001.
11. C1 emptied every batch: U-link (DBB) + P-101 residual to C2 via HV-120 (M-17 / M-20).
12. Gallery floor falls 1:100 to the sump; P-105 auto, discharge riser to grade + blind, destination HOLD.

LEGEND

- Raw / wet oil (P-103 suction & return)
- Sludge
- Water
- Pump discharge riser
- Pipe turning up (riser)
- Pipe turning down / bell
- Concrete (cut)
- Hatch / opening above

SEVERO WM OIL PROJECT

PS-01 PIT + DRY GALLERY — PLAN

Doc. No.: SEVERO-WMO-CL-ENG01-PS01-LAY-001

Rev. 2 — IFR (Issued for Review)

supersedes LAY-001 Rev.1 (all sheets)

Class CL — Client Deliverable

Date: 19-Oct-2026 Author: Eng. Mohamed Tarek

Ref.: PS01-DB-001 Rev.2; GEN-PID-001 Rev P1 Sh. 03/03A/04

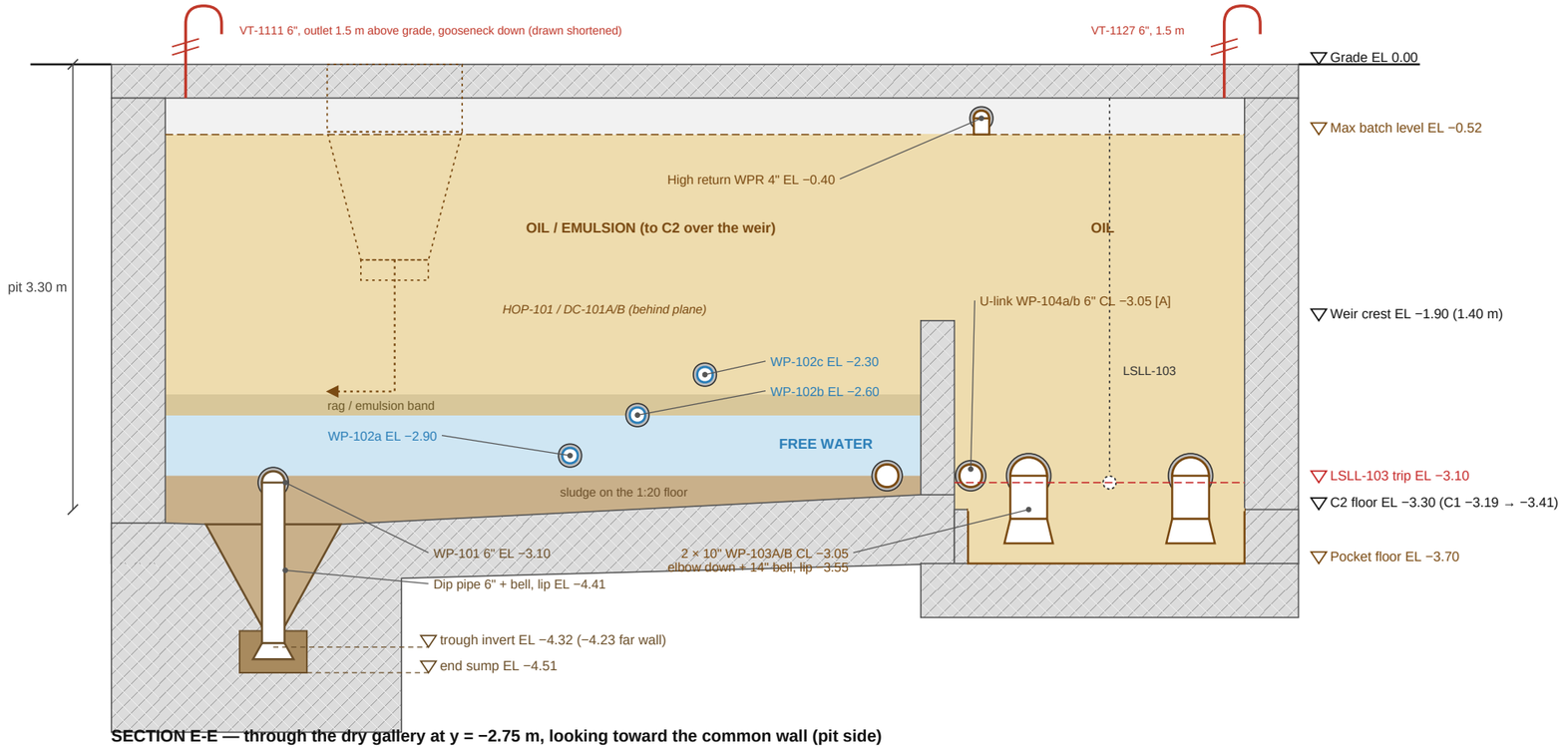
Sheet 1 of 3 A3 — NTS (drawn to scale ≈ ratio)

PS-01 — RECEPTION PIT + DRY PUMP GALLERY — SHEET 2: LONGITUDINAL SECTIONS A-A AND E-E

CONCEPT / IFR — drawn to scale ratio (NTS). Same horizontal scale as Sheet 1: wall pieces line up vertically between the two views. EL in m from grade 0.00.

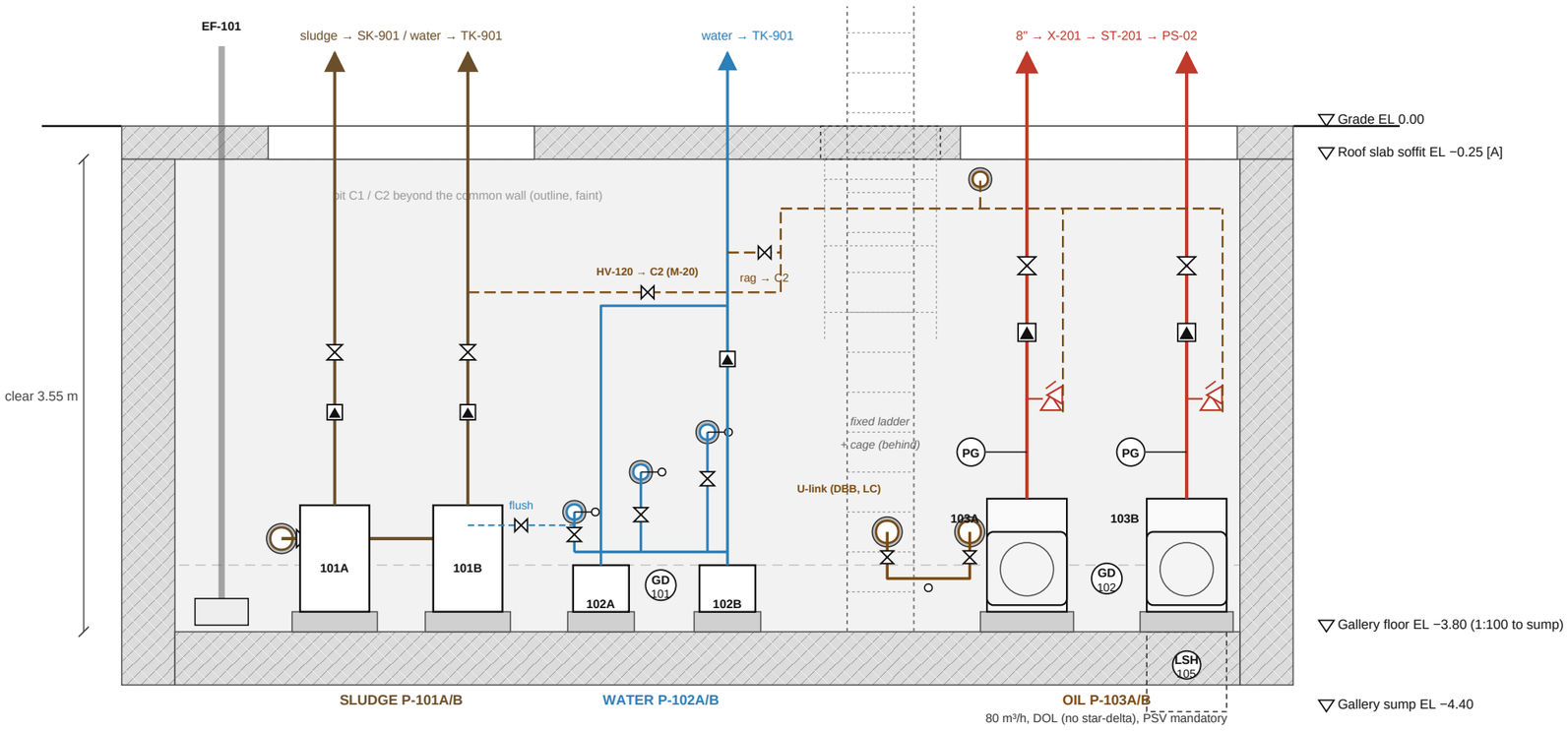
SECTION A-A — through the pit at y = 0.62 m, looking toward the common wall (gallery side)

Liquid layers shown are ILLUSTRATIVE (end of settling, before draw-off). Items behind the section plane shown dotted.



SECTION E-E — through the dry gallery at y = −2.75 m, looking toward the common wall (pit side)

Wall pieces seen on the common wall face. The plane cuts the P-103 motors (pumps stand at 90° to the wall; heads, valves, strainers behind).



KEY ELEVATIONS (from grade 0.00)

Max batch level (C1 emptied)	**−0.52**
High return WPR (4")	**−0.40**
Weir crest (1.40 m)	**−1.90**
Water draw-offs WP-102a/b/c	**−2.90 / −2.60 / −2.30**
P-103 wall pieces 10" (CL)	**−3.05**
LSLL-103 trip	**−3.10**
WP-101 6" / U-link WP-104a/b 6" (CL)	**−3.10 / −3.05**
C1 floor weir → trough / C2 floor	**−3.19 → −3.41 / −3.30**
P-103 bell lip / pocket floor	**−3.55 / −3.70**
Gallery floor	**−3.80**
Trough invert / end sump	**−4.23 → −4.32 / −4.51**
Sludge bell lip	**−4.41**

NOTES

1. Every discharge riser: PG, NRV, gate valve; P-103 also PSV ≈ 5.5 bar(g) → overhead return → C2 (mandatory).
2. P-103A/B: 80 m³/h, fixed speed, DOL start — NOT star-delta (full torque needed with cold oil).
3. Pumps flooded: liquid in pit always above pump inlets during operation (order sludge → water → oil).
4. Sludge suction: one 6" dip pipe + header to 1+1 pumps; water flush connection to clear blockages.
5. Sample cock on each water wall piece (upstream of its valve) shows oil / emulsion / water at that level.
6. Gallery 8.0 × 3.5 m [A] (= pit length); access by fixed ladder + cage (ISO 14122-4) through a roof hatch.
7. Footprints, motor sizes, slab thicknesses [A].
8. Layers in A-A are illustrative only.
9. Rev.2: U-link (DBB) at the foot of the weir; P-101 residual → C2 via HV-120; vents 1.5 m, gooseneck down; hopper = collar 1700 × 1000 + pyramid (M-04).

LEGEND

- Oil
- Sludge
- Water
- P-103 discharge
- Cast-in wall piece with puddle flange
- Gate valve / NRV

SEVERO WM OIL PROJECT

PS-01 PIT + DRY GALLERY — SECTIONS A-A, E-E

Doc. No.: SEVERO-WMO-CL-ENG01-PS01-LAY-001

Rev. 2 — IFR (Issued for Review)

supersedes LAY-001 Rev.1 (all sheets)

Class CL — Client Deliverable

Date: 19-Oct-2026 Author: Eng. Mohamed Tarek

Ref.: PS01-DB-001 Rev.2; GEN-PID-001 Rev P1 Sh. 03/03A/04

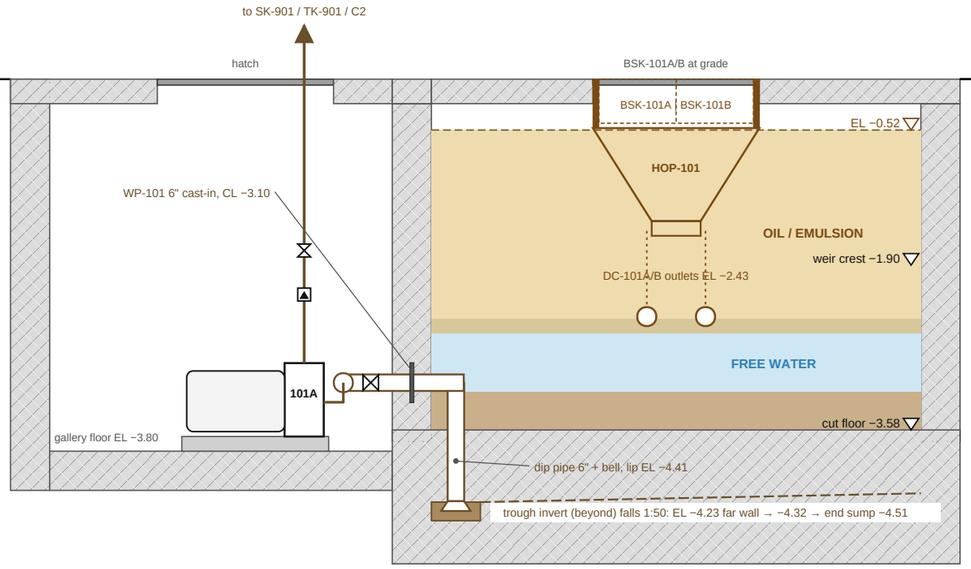
Sheet 2 of 3 A3 — NTS (drawn to scale ≈ ratio)

PS-01 — RECEPTION PIT + DRY PUMP GALLERY — SHEET 3: TRANSVERSE SECTIONS B-B, C-C, D-D

CONCEPT / IFR — Rev.2 — drawn to scale ratio (NTS). All three sections look toward the inlet end (−x): gallery on the left, pit on the right. EL in m from grade 0.00.

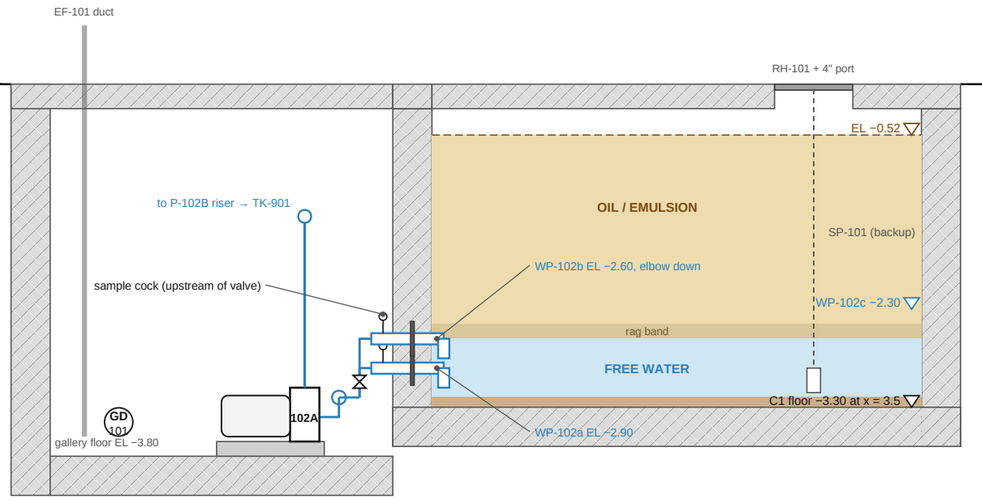
SECTION B-B (x = 1.2 m) — sludge trough, dip pipe, P-101A

cut through the trough side slope, the hopper and P-101A



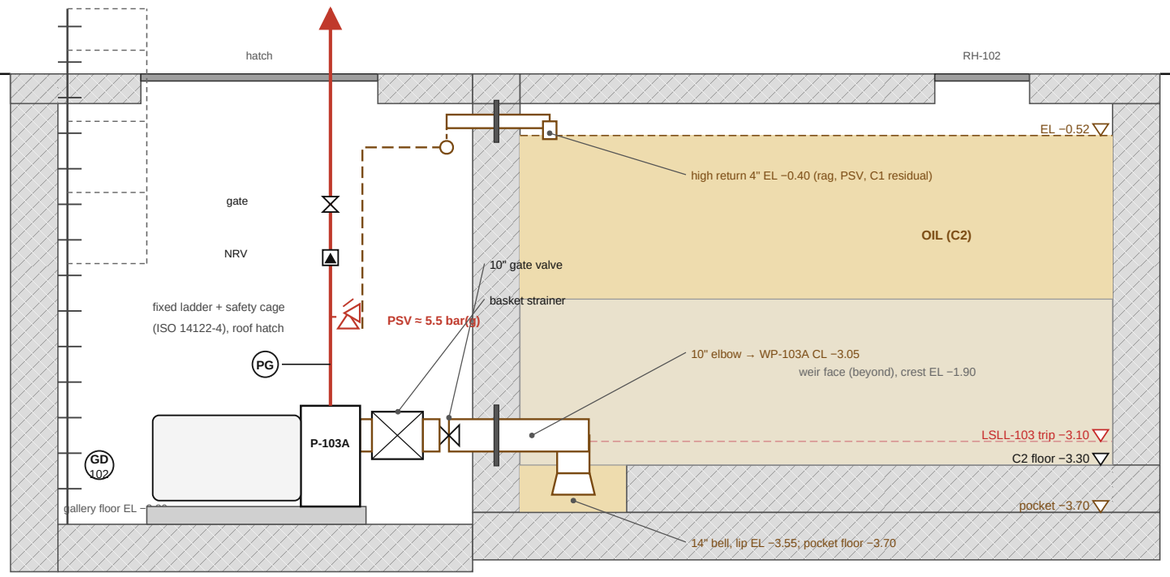
SECTION C-C (x = 3.5 m) — water draw-offs + sample cocks, P-102A

WP-102b on the plane, WP-102a in view (WP-102c is behind the viewer)



SECTION D-D (x = 6.4 m) — C2 suction pocket, bell, WP-103A, P-103A

flooded suction; P-103A stands at 90° to the wall (head, then gear-motor toward the walkway)



HOW THE PIT IS OPERATED — ONE BATCH (Rev.2, M-12 / M-13 / M-17 / M-20)

0. Permissives: C1 emptied last batch; U-link HV-115/116 locked closed; P-103 stopped.
1. RECEIVE ≤ 75 m³/h (BSK-101A/B → HOP-101 → DC-101A/B). P-102 runs from WP-102a for the whole unloading; tanker 2 only when the WP-102b cock shows oil; water at WP-102c or LAH-101 → stop unloading. P-103 never runs while unloading.
2. SETTLE; read the layers at the WP-102a/b/c cocks (SP-101 backup).
3. SLUDGE: P-101 → skip (HV-119); water at SP-103 → HV-118 → TK-901; stop at oil.
4. WATER: P-102 from the lowest draw-off showing water → TK-901; throttle HV-110.
5. RAG (when present): P-102 → HV-112 → WPR → C2.
6. U-LINK: open HV-115, crack SP-102: oil → open HV-116 (water → back to step 4).
7. OIL: P-103 → X-201 → ST-201 → PS-02, C1 + C2 together to LSLL-103 (EL −3.10).
8. C1 RESIDUAL ≈ 7.8 m³: P-101 → HV-120 → WPR → C2; P-103 again to LSLL-103.
9. Close HV-116 then HV-115 (LC); no flow at SP-102. Ready (cycle ≈ 5.9 h).

NOTES

1. Sections look toward −x (inlet end): gallery left, pit right; things nearer the inlet are in view.
2. Every wall piece is cast-in with a puddle flange (dark bar inside the common wall).
3. P-103 suction per pump: bell → 10" elbow → wall piece → 10" gate valve → basket strainer → pump.
4. P-103 discharge: PG → PSV ≈ 5.5 bar(g) (to C2) → NRV → gate valve → 8" riser. Fixed speed, DOL.
5. Sludge: 6" dip pipe from the end sump, header to P-101A/B (1+1), no suction strainer, flush line.
6. Water: sample cock on each wall piece upstream of its valve; P-102 horizontal centrifugal 1+1.
7. Footprints, slab thicknesses, motor sizes [A].
8. Liquid layers illustrative only.
9. Rev.2: weir 1.40 m (crest −1.90); C1 floor 1:20; trough invert 1:50 to sump −4.51; draw-offs with elbows down; LSLL-103 at −3.10. HOP-101 = collar 1700 × 1000 to −0.50 + pyramid to 500 × 500 at −1.45.

LEGEND

- Oil
- Sludge
- Water
- P-103 discharge
- Puddle flange (cast in wall)
- Concrete (cut)

SEVERO WM OIL PROJECT

PS-01 PIT + DRY GALLERY — SECTIONS B-B, C-C, D-D

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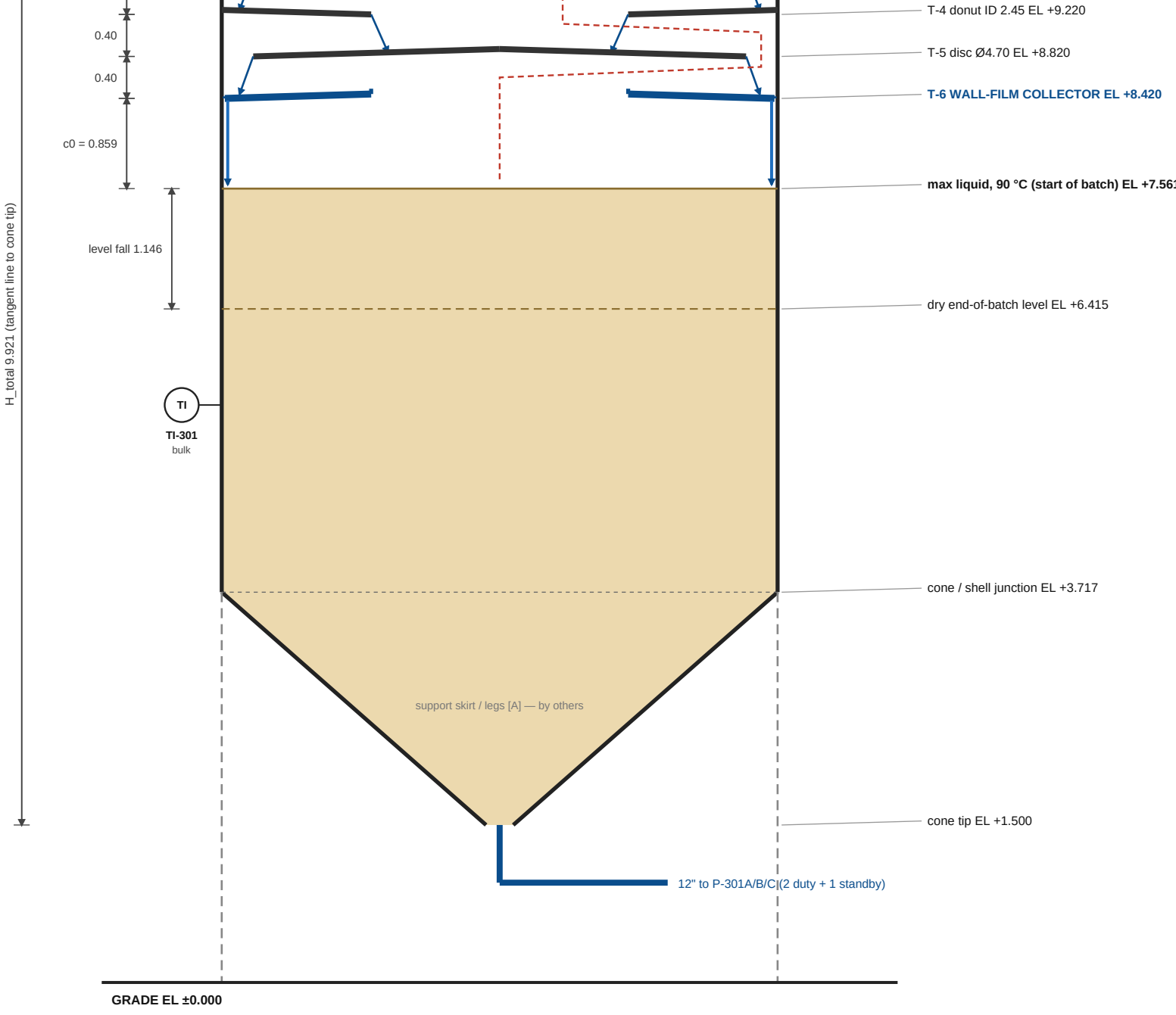
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Sheet 3 of 3 A3 — NTS (drawn to scale ≈ ratio)



Legend: oil (curtains, films) vapour (indicative)